

Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning
Graduate School of Architecture, Planning and Preservation
Columbia University

April 12, 2026

Abstract

This thesis asks whether formal protest petitions against housing development proposals in Austin, Texas can be ranked by risk before formal public hearings, using only information available at the time of filing. The outcome is a *measured threshold-crossing petition*—an operationalized indicator of whether the reconstructed petition-share calculation exceeded the statutory 20% threshold in public records. Because municipal clerk certification is not directly observed, this measure is distinct from a legally certified valid petition. Working from 7,074 discretionary zoning cases filed under Austin’s pre-HB 24 regime (2007–2024), out-of-distribution bootstrap evaluation at the filing-date horizon yields a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]), with an Expected Calibration Error of 0.108 after Isotonic Regression—above the 0.05 threshold for reliable probabilistic interpretation. Filing-date administrative features can rank petition risk, but they do not yet yield portable, well-calibrated probabilities for administrative deployment.

Acknowledgments

I would like to express my deepest gratitude to my thesis advisor, Dory Kornfeld, for her guidance, support, and expertise throughout this research. Her insights into urban planning and policy were invaluable in shaping this work.

I am profoundly grateful to my thesis readers for their time, expertise, and thoughtful feedback: Matthew Bauer, Adjunct Associate Professor; Josh Begley, Adjunct Assistant Professor; James Piacentini, Adjunct Assistant Professor; Stijn Van Nieuwerburgh, Earle W. Kazis and Benjamin Schore Professor of Real Estate and Co-Director of the Paul Milstein Center for Real Estate at Columbia Business School; and Adam Vosburgh, Associate Director and Adjunct Assistant Professor. Their interdisciplinary perspectives spanning computational design, real estate finance, and urban planning enriched this research immeasurably.

I would like to thank Professor Weiping Wu, Vice Provost for Academic Programs and Director of the Urban Planning Program at Columbia University, for her leadership of the program and support of my academic journey.

I am grateful to the Columbia University Graduate School of Architecture, Planning and Preservation (GSAPP) for providing the academic environment and resources that made this research possible.

Finally, I thank my colleagues, peers, and the broader Columbia community for their intellectual engagement, encouragement, and support throughout this journey.

Contents

1	Introduction	7
2	Literature Review	8
2.1	The Political Economy of Zoning Friction and Participatory Distortion	8
2.2	Discretionary Review and Institutional Vetoes	8
2.3	The Prediction Policy Problem in Spatial Planning	11
2.4	Evaluative Constraints on Predictive Governance	11
3	Data and Methods	11
3.1	Analytical Panel Structure	12
3.2	Case Universe and Sample Selection	14
3.3	Primary Outcome Definition	16
3.4	Feature Libraries: Policy Forecasting vs. Explanatory Research	16
4	Modeling Strategy	17
4.1	Supporting Stages: Development Occurrence, Project Type, and Political Outcome .	17
4.2	Formal Protest Petition	17
4.3	Model Evaluation Strategy	19
5	Results	20
5.1	Predictive Performance Across All Horizons	20
5.2	Predictors of Petition Filing	22
5.2.1	Attribution Stability Across Temporal Windows	24
5.3	Institutional Context: Regression Discontinuity at the 20% Petition Threshold . . .	25
5.4	Descriptive Evidence from the 2022 Electoral Transition	28
5.5	Supplementary Institutional Analyses	28
6	Qualitative Planning Context	29
7	Limitations	30
8	Conclusion	31
	References	33
A	Supplementary Calibration Comparison	36
B	Semantic Feature Mapping	37
C	Interview Recruitment Overview	38
D	IRB Protocol Summary	39
E	Semi-Structured Interview Guide	40
F	Sample Protest Petition	41
G	Illustrative Support-Layer Panel Structure	42

H	Methodological Appendix: Expanded Details	43
H.1	Variable Dictionary	43
H.2	Hyperparameter Search Grids	43
H.3	Subgroup Definitions & Fairness Testing	44
H.4	Natural Language Processing Model: Exploratory Text Features	44
I	Development Occurrence Model (Stage A)	47
J	Project Type and Scale Model (Stage B)	50
K	Political Outcome Model (Stage D)	51
L	Expected Contested Units (Composite Measure)	52
M	Institutional Geographic Overlay Analyses	53
N	HOME Initiative Event Study	55
O	Invariant Causal Prediction (ICP)	56

List of Figures

1	Annual Petition Filing Rate by Project Type	10
2	Spatial Distribution of Zoning Cases (2007-2024)	13
3	200ft Parcel Buffer Geometries	14
4	Schematic of the Austin Zoning Process	15
5	Formal Protest Petition PR Curves by Model	18
6	Calibration of the Stage A and Stage C Models	22
7	Performance Across Temporal Holdouts	23
8	Top Predictor Groups After Hierarchical Clustering	23
9	SHAP Beeswarm Plot for the Opposition Model	24
12	Causal Identification: Regression Discontinuity DAG	25
10	Expanding-Window Attribution Stability Test	26
11	Multi-Horizon Attribution Stability	27
13	Threshold-Based Discontinuity at the Measured 20% Petition Share	28
14	Geographic Distribution of False Positives vs. Actual Filings	31
15	Supplementary Hyperparameter Search Results	43
16	Mean Predicted Probability of Argument Frames by District	45
17	Argument Frame Probabilities: Opposed vs. Unopposed Cases	46
18	Development Hazard Classification PR Curves	48
19	Predicted Geographic Occurrence vs. Realized Events	49
20	Composite Measure of Expected Petitioned Units	52
21	Estimated 2022 Electoral-Transition Interaction Effects by District	53
22	Placebo Estimates Across Earlier Election Cycles	54
23	HOME Phase 1 Event-Study Coefficients	55

List of Tables

1	Measured Threshold-Crossing Petition Incidence by Proposed Land Use in Austin (Mercatus Center)	9
2	Annual Distribution of Discretionary Zoning Cases (2007—2024)	13
3	Historical Panel Descriptive Statistics	15
4	Sample Selection from Raw Records to Analytic Sample	15
5	Precision-Recall Area Under Curve (PR-AUC) Evaluation	18
6	Development-proposal and petition-filing models evaluated at filing: Raw vs. calibrated candidate models	19
7	Filing-Date Predictive Performance for the Petition Model	21
8	Multi-Horizon Petition-Filing Model Performance with 95% Bootstrap CIs	22
9	Calibration Method Comparison: ECE, ACE, and Brier Score Across Models	36
10	Semantic Feature Aggregation Mapping	37
11	Stage A Target Definitions	47
12	Development Occurrence Hazard Model Performance	48
13	Project Type and Scale	50
14	Pre-Council Attrition Forecast (Opposed Cases Only)	51
15	Summary of Supplementary Quasi-Experimental Estimates	54

1 Introduction

As the United States confronts a persistent housing affordability crisis, understanding local barriers to new housing supply remains an important empirical challenge. Traditional accounts emphasize the role of municipal zoning laws, but a substantial share of delay and blockage occurs through discretionary political processes—public hearings, protest petitions, and council negotiations—that collectively slow or block individual projects. This thesis uses Austin, Texas as a case study. Between 2010 and 2024, rapid employment growth drove severe housing supply constraints, inflating property values and generating sustained conflict between neighborhood preservation and regional development goals. The study is historically bounded to Austin’s pre-HB 24 protest petition regime (2007–2024) and serves as a template for analyzing institutional change in local land-use governance, not as a near-term operational tool under current Texas law.

Against this backdrop, the central research question is whether a measured threshold-crossing petition—defined below in Section 3.3—can be predicted before formal public hearings begin.

The outcome under study is specific. Under the pre-2025 framework of Texas Local Government Code § 211.006 [29], adjacent property owners may file a protest petition; if signatures representing 20% of the relevant adjoining land area were collected and the municipal clerk certified the petition as valid, the petition historically required a three-fourths supermajority for City Council approval. (Effective September 1, 2025, House Bill 24 repealed and replaced this mechanism with a differentiated, multi-tier system under § 211.0061.) This is one institutionalized form of neighborhood opposition—the most consequential and best-observed form in the Austin regulatory context—not a prediction of opposition in all its forms. The thesis narrows its claim accordingly: the primary contribution is an assessment of whether administrative, spatial, and economic features observable at the time of a zoning application are sufficient to anticipate this specific institutional event.

To answer this question, the research constructs a sequential predictive methodology organized around four stages:

- **Stage A—Development Occurrence:** Where housing development proposals are likely to emerge, modeled as a discrete-time site hazard across 282,000 candidate parcels.
- **Stage B—Project Type and Scale:** Conditional on a development event, the expected project form and unit count.
- **Stage C—Formal Protest Petition:** Conditional on a project entering the formal public process, the probability that a measured threshold-crossing petition is filed. *This is the core of the thesis.*
- **Stage D—Political Outcome:** Conditional on project characteristics and opposition status, the likelihood of pre-council withdrawal or final disposition.

Stages A, B, and D provide context for Stage C. Stage A supplies inverse-probability weights to adjust for spatial selection into the analytic sample. Stage B characterizes the heterogeneous institutional responses associated with different project types. Stage D documents the downstream consequences of petition filing, including high rates of pre-council withdrawal among opposed cases. The predictive evaluation, however, centers on Stage C: whether the model can discriminate petition risk at the filing-date horizon and whether it does so with sufficient calibration and equity to support high-confidence administrative use.

The thesis also includes two exploratory analyses of downstream institutional dynamics. A regression discontinuity design at the 20% petition threshold estimates the processing-time consequence

of crossing the statutory margin. Descriptive evidence from the 2022 council electoral transition, in which specific districts replaced preservationist incumbents with pro-housing representatives, examines whether petition filing rates declined differentially in those districts.

Finally, semi-structured interviews with planning professionals, conducted under IRB Protocol ACYY0820, informed the interpretation of quantitative findings by surfacing institutional dynamics—informal negotiations, strategic petition timing, neighborhood network mobilization—that administrative data cannot directly observe. The interviews do not validate the model; they contextualize findings within professional planning practice and surface governance concerns about potential misuse of predictive outputs.

2 Literature Review

This literature review synthesizes scholarship across three core themes to directly motivate the thesis’s methodological approach. It first establishes the political economy of homeowner opposition and participatory distortion. Second, it isolates the specific institutional mechanism of the protest petition. Finally, it provides the intellectual justification for treating zoning conflict as primarily a predictive policy problem while acknowledging the inherent evaluative constraints required for administrative use of predictive models in spatial planning.

2.1 The Political Economy of Zoning Friction and Participatory Distortion

The foundational context for this research rests on the consensus that land-use regulation functions as a binding constraint on housing supply, driving a wedge between fundamental construction costs and market prices [14, 13]. While macro-economic literature establishes the downstream effects of this regulatory tax—including suppressed economic growth and diminished housing affordability—urban planning scholarship points to the local political economy to explain why such constraints persist despite citywide needs [15].

The theoretical centerpiece for understanding this localized resistance is Fischel’s [12] homevoter hypothesis. Fischel posits that homeowners engage in local politics primarily to protect both the use value and exchange value of their homes, rationally utilizing zoning restrictions to preserve neighborhood exclusivity and mitigate perceived financial risks. However, recent empirical work has demonstrated that this protective impulse translates into a highly asymmetric political arena. Einstein, Palmer, and Glick [11] document the phenomenon of “participatory distortion,” demonstrating that the participants who dominate planning and zoning board meetings are overwhelmingly older, disproportionately white, longtime homeowners who are united in opposing new housing. These “neighborhood defenders” possess the time, resources, and localized knowledge to effectively block individual projects. Furthermore, Hankinson [17] complicates the pure asset-protection motive by revealing that renters in high-cost cities exhibit spatial NIMBYism on par with homeowners due to localized displacement anxiety, cementing the reality that spatial proximity to development triggers mobilization regardless of tenure.

2.2 Discretionary Review and Institutional Vetoes

Participatory distortion matters for housing supply when it operates through institutions that can delay or block projects. Manville and Monkkonen [22] highlight that discretionary review processes—where development relies on site-specific municipal approvals rather than by-right administrative clearance—serve as the primary vehicle through which local opposition exerts leverage. In a

discretionary regime, citywide housing goals are repeatedly displaced by parcel-level political conflict [19].

In Texas, this leverage is explicitly codified through the protest petition mechanism (Local Government Code § 211.006). The statute legally empowers adjacent property owners to force supermajority city council voting requirements if owners representing 20% of the *area of lots* immediately adjoining the proposed change and extending 200 feet from that area sign a valid petition.¹ This mechanism empowers localized opposition, transforming protest into a formal institutional hurdle. Cellini, Ferreira, and Rothstein [7], in their analysis of California school bond referenda, demonstrate how supermajority thresholds inherently distort political outcomes to favor the status quo.

In Austin, the protest petition creates institutional friction for land use: cases exceeding the measured petition threshold appear to face additional voting requirements and longer processing times. Recent empirical research by the Mercatus Center at George Mason University confirms the severity and asymmetry of this mechanism specifically within the Austin political geography. Their analysis demonstrates that valid petitions are more common for rezonings that increase residential density: historically, 25% of rezonings to multifamily use in Austin faced a valid petition, compared to only 5% for commercial rezonings.

Table 1: Measured Threshold-Crossing Petition Incidence by Proposed Land Use in Austin (Mercatus Center)

Proposed Zoning Category	Rezonings Facing Valid Petition (%)
Multifamily Residential	25%
Commercial / Non-Residential	5%

¹§ 211.006(d) and (f) were repealed effective September 1, 2025 by HB 24 (89th Texas Legislature). The operative provisions now reside in new § 211.0061, which establishes a three-tier protest system: (1) owners of $\geq 20\%$ of the directly affected area trigger a three-fourths vote; (2) owners of $\geq 60\%$ of the adjoining 200-foot buffer trigger a three-fourths vote; and (3) for qualifying residential development under subsection (b)(3), a simple majority of all governing body members is required. Proposed comprehensive zoning changes are exempted from the protest mechanism entirely.

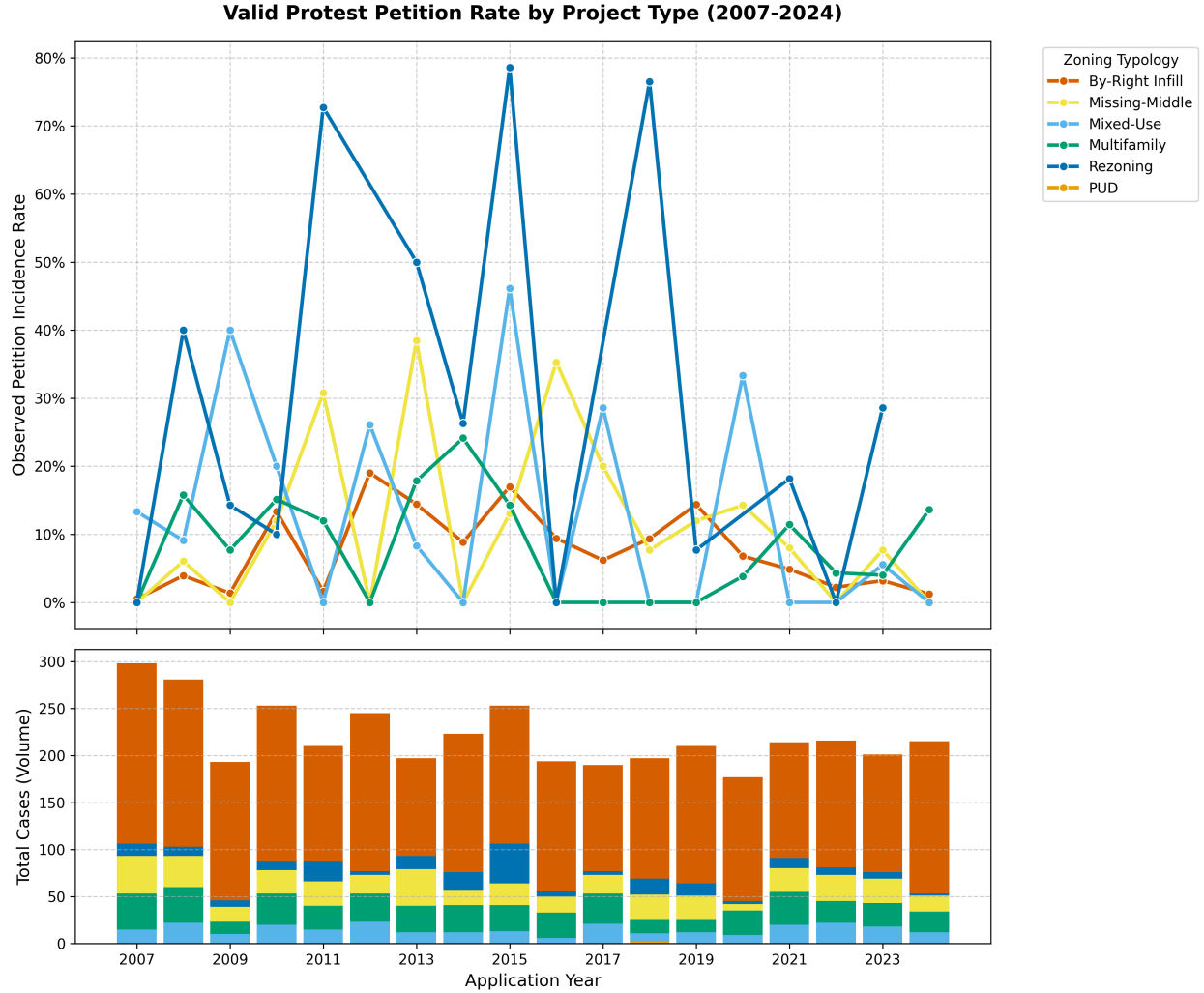


Figure 1: Annual petition filing rates by project type, 2007–2024. Consistent with the Mercatus Center pattern (Table 1), this timeline tracks the observed historical petition frequency within each development category over the study period. While high-density or specialized use categories historically experience greater petition volatility, missing-middle and standard residential rezonings—which comprise the vast majority of all discretionary cases—maintain a consistently low baseline filing rate throughout the dataset.

Because the process is fundamentally asymmetric—empowering opponents of development through a fixed statutory buffer geometry—it creates a bias toward the status quo, dragging contested housing applications into months of procedural delays. The Texas Legislature structurally altered this framework in 2025; under House Bill 24, the state established new § 211.0061, which exempts proposed comprehensive zoning changes from the protest mechanism entirely and replaces the uniform 20% trigger with a differentiated, multi-tier system. For qualifying residential protests under subsection (b)(3), the vote consequence is a simple majority of all governing body members rather than the prior three-fourths supermajority; for other protest paths, three-fourths remains the threshold. Consequently, predicting whether a measured threshold-crossing petition will be filed against an application under the pre-HB 24 regime forms the core empirical challenge of this thesis.

2.3 The Prediction Policy Problem in Spatial Planning

Traditional econometric scholarship in urban planning almost exclusively evaluates the causal effect of zoning—for instance, whether upzoning increases land values or whether a protest petition causes delay. However, addressing the localized gridlock of discretionary review requires anticipating *where* opposition will emerge before political capital is expended. This intellectual justification builds on Shmueli’s [28] foundational distinction between explanatory modeling (testing causal hypotheses) and predictive modeling (generating accurate forecasts of new observations), and on Mullainathan and Spiess [23], who argue that machine learning solves a fundamentally different problem than traditional econometrics: prediction ($\hat{y}$) rather than parameter estimation ($\hat{\beta}$).

Kleinberg, Ludwig, Mullainathan, and Obermeyer [21] formalize this distinction by defining the “predictive policy question,” cases where optimal policy decisions depend not on understanding the causal mechanism of an outcome, but on accurately forecasting its occurrence. Predicting which zoning cases will meet the petition threshold can be framed as a predictive policy question. Planners and policymakers need to know which projects will trigger supermajority voting requirements to efficiently allocate political capital, engage in proactive community mediation, or design comprehensive reforms that bypass parcel-level friction [2]. By applying algorithmic forecasting to administrative municipal data, this research transitions the study of NIMBYism from a retrospective causal analysis into a proactive predictive model, testing whether administrative spatial characteristics carry sufficient signal to anticipate formal protest petitions in formal proceedings weeks or months before a public hearing takes place.

2.4 Evaluative Constraints on Predictive Governance

While transitioning to a predictive methodology offers potential analytical value for municipal planning, it introduces serious ethical risks. Barocas and Selbst [3] demonstrate that predictive algorithms inevitably inherit the prejudices embedded in their training data. Because historical zoning data in American cities is inextricably linked to legacies of redlining, racial covenants, and exclusionary zoning [31], the underlying spatial features, such as lot geometry, single-family zoning, and historic property values, function as proxies for race and class. If an algorithm learns that wealthy, historically white neighborhoods successfully file protest petitions, using that prediction to guide development decisions risks encoding a feedback loop that shields those neighborhoods from densification while concentrating development pressure on politically vulnerable communities [27].

Consequently, the viability of algorithmic prediction in spatial planning is constrained by mathematical limits on fairness. Chouldechova [9] proves that when base rates of an outcome differ across groups, no imperfect predictor can simultaneously satisfy both predictive parity and classification parity. Recognizing these fixed mathematical constraints, modern predictive models intended for public-sector use must be evaluated against explicit calibration and error-rate thresholds. Rather than attempting to maximize pure predictive lift, the administrative application of such models should consider explicit calibration accuracy and the equalization of False Negative Rates across geographic groups, ensuring that predictive efficiency does not come at the cost of democratic equity.

3 Data and Methods

The empirical foundation is a time-stamped spatial panel dataset that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date,

ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

3.1 Analytical Panel Structure

The core analytics rely on the filing-date analytic dataset, which merges spatial and demographic variables into single case-level observations (with temporally matched covariates):

1. **Zoning Application Records:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status.
2. **Spatial Geometries:** Parcel IDs, geometries, acreage, frontage, and 200ft buffers used to aggregate nearby properties (council districts and overlapping geometries).
3. **Parcel Appraisals:** Annually indexed TCAD appraisal variables for parcels used to compute variables within the 200-foot buffer: land value, structure age, homestead exemption rates, and owner-occupancy shares.
4. **Neighborhood Demographics:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators.
5. **Institutional Calendar:** The 2022 council regime change; the HB 24 effective date (September 1, 2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, precise milestone dates are reconstructed through two methods. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using the minimum procedural constraints mandated by Texas Local Government Code and Austin municipal zoning ordinances, such as statutory 11-day mail notice minimums.

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

Annual Zoning Application Volume (2007-2024)

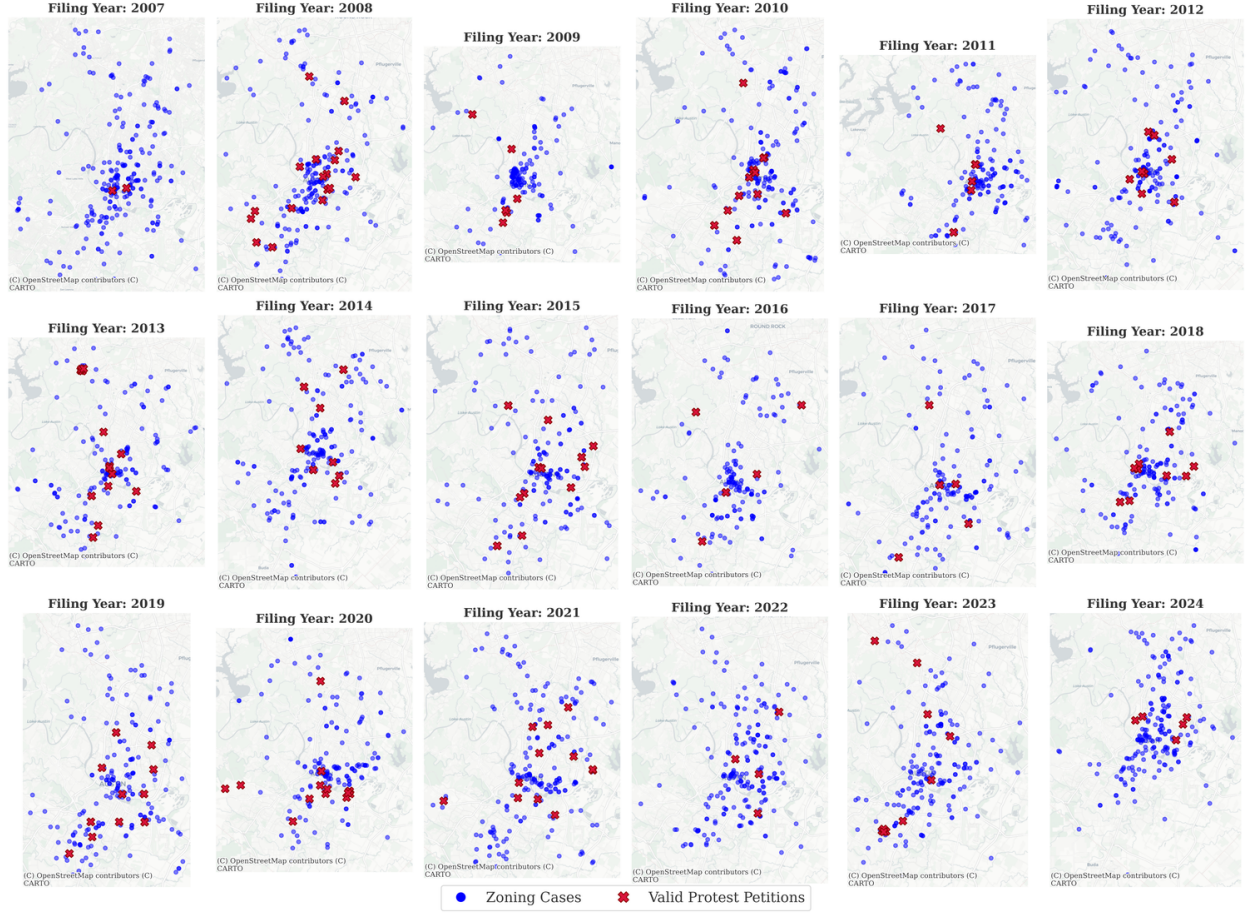


Figure 2: Spatial distribution of zoning cases across the Austin municipal core ($N = 7,074$).

Table 2: Annual Distribution of Discretionary Zoning Cases (2007—2024)

Year	Cases	Year	Cases	Year	Cases
2007	288	2013	366	2019	460
2008	270	2014	390	2020	362
2009	265	2015	454	2021	476
2010	297	2016	422	2022	436
2011	369	2017	483	2023	625
2012	335	2018	492	2024	284
Total (2007—2024): 7,074 cases					

**Visualizing the 200ft Statutory Protest Buffer
(Empirical TCAD Parcel Boundaries for Case C14-2007-0131)**

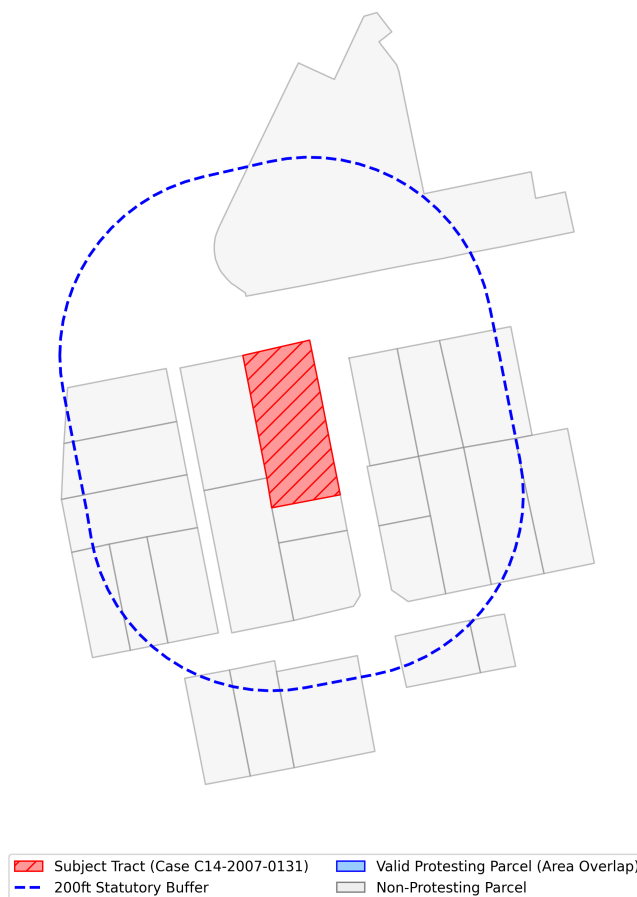


Figure 3: 200ft parcel buffer geometries used for TCAD appraisal aggregation and defining the petition buffer.

3.2 Case Universe and Sample Selection

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances, such as 5-foot setback adjustments. Because the petition mechanism under study—formal protest petitions under Texas LGC Chapter 211—does not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the legal protest-petition mechanism, these cases are excluded.

Figure 4 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

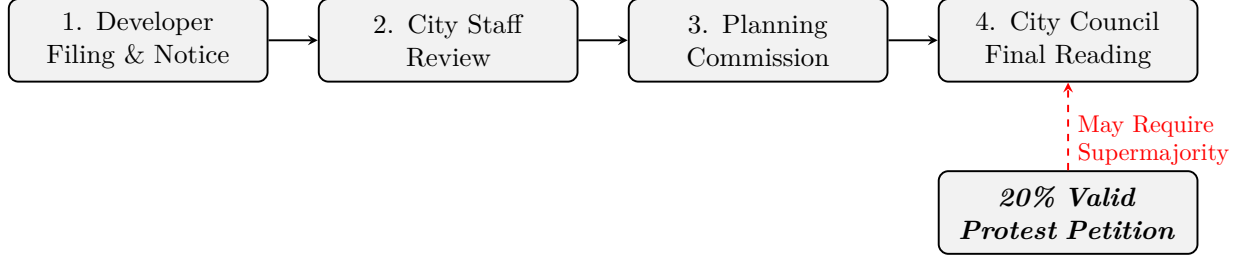


Figure 4: Austin municipal zoning process. The figure schematically shows where a qualifying protest petition could make a three-fourths council vote relevant under the study-period framework. A petition representing 20% of the area of lots immediately adjoining the proposed change would require, under the applicable legal framework, a supermajority vote of the City Council for approval, though recusal or absence of council members can alter the exact number of affirmative votes needed.

Table 3: Historical Panel Descriptive Statistics

	count	mean	std	min	Median	max
Gross Site Area (Acres)	534	4.92	30.66	0.00	0.00	669.83
Requested Height Delta (ft)	3994	10.5	34.9	-60	0	365
Requested FAR Delta	3994	0.43	0.95	-3.00	0.05	7.60
Requested Bldg Cov Delta (%)	3994	12.3	20.7	-55	5	80
Filing Year	7074	2008.6	10.11	1966	2009	2026
Organized Opposition (Binary)	7153	0.06	0.23	0	0	1

Table 4 reports the sample selection from raw administrative records to the analytic sample.

Table 4: Sample Selection from Raw Records to Analytic Sample

Selection Step	Cases Remaining
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Restrict to discretionary cases (rezonings, PUDs, NPAs)	7,153
Exclude cases with unusable or missing initial filing-year information	7,074
Final Analytic Sample (Stages C/D)	7,074

The analytical samples vary by analytical goal. The primary predictive evaluation (Stage C) uses the full sample of 7,074 discretionary geographic cases. In contrast, the supporting analyses use restricted sub-samples designed to isolate specific mechanisms. The 2022 electoral-transition descriptive analysis and the HOME event study utilize $N = 518$ because they are restricted to active single-member districts immediately surrounding the relevant election or policy cycle. The Invariant Causal Prediction (ICP) analysis utilizes $N = 1,186$ multi-parcel geometric environments to represent cohesive spatial clusters. Finally, the institutional geographic overlay regressions evaluate policy flags aggregated at the council-term level across periods, resulting in $N = 20$ macro-observations.

Unlike approaches that require complete-case geometric intersections, this methodology uses CatBoost’s native missing-value handling to build splits over absent variables, such as unmatched TCAD boundaries, without listwise deletion. This approach retains cases with partially missing covariates, reducing artificial attrition and the selection bias introduced by demanding full covariate coverage. Whether it also improves fairness metrics is a separate empirical question explored in Section 7.

3.3 Primary Outcome Definition

The primary predictive outcome (y_0) is a *measured threshold-crossing petition*: a binary indicator equal to one when the reconstructed petition-share calculation—derived from public petition filings and buffer-area computations—exceeds the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284. This is an operationalized outcome, not a direct observation of the legally decisive event. The City Clerk’s formal certification of each petition as legally valid is not recorded in the public dataset; consequently, the outcome captures cases that *appear* to have crossed the threshold based on available records, which may include petitions that were ultimately rejected on procedural grounds or petitions whose true area share differed from the calculated value. Throughout this manuscript, “threshold-crossing petition” refers to this reconstructed measure, and “legally certified petition” is reserved for the unobserved clerk-certified outcome.

Secondary outcomes are analyzed separately:

- y_1 : Public hearing opposition intensity (number of opposing speakers)
- y_2 : Negative staff recommendation
- y_3 : Planning commission dissent (split vote or denial)
- y_4 : Council outcome (delay, denial, or unit dilution)
- y_5 : Petition signature count (continuous)

All primary results report y_0 . Sensitivity analyses replicate key findings using individual secondary outcomes. This separation is important because staff recommendations, commission votes, and council outcomes reflect different institutional mechanisms than neighborhood mobilization through petitions, and collapsing them into a single binary would obscure meaningful distinctions. Cases withdrawn before public notice are excluded, rather than coded as zero, because no opportunity for opposition existed. ETJ cases and routine variances are excluded because the formal petition mechanism does not apply to them.

Note on stage-specific samples: the Stage C filing-date model uses the full 7,074-case analytic sample regardless of whether a hearing transcript exists, since transcripts post-date the filing horizon and are not used as predictive features at that stage. Cases missing hearing transcripts or petition-database coverage are omitted only from later-horizon evaluations and secondary-outcome analyses where those records serve as inputs or outcome sources.

3.4 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is separated to prevent predictive models from leveraging sensitive attributes:

Predictive features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators

(median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates). These elements are selected strictly based on their availability in public administrative and appraisal databases prior to the initial filing date, ensuring the algorithm’s decisions replicate constraints faced by municipal planning staff in real time.

Explanatory research features are restricted to equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification. These features are strictly partitioned from the predictive model to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 7). By reserving these variables exclusively for post-hoc fairness evaluations, the model avoids generating predictions based on protected class attributes while maintaining the capability to measure intersectional disparities across geographic subgroups.

4 Modeling Strategy

4.1 Supporting Stages: Development Occurrence, Project Type, and Political Outcome

The predictive evaluation centers on Stage C (Formal Protest Petition). Three supporting stages provide context and methodological inputs:

Stage A—Development Occurrence models where housing development proposals are likely to emerge as a discrete-time site hazard across 282,000 candidate parcels. Its primary methodological role is to supply inverse-probability weights (IPW) that adjust for spatial selection into the analytic sample: the filing-date petition model observes only neighborhoods that actually received development proposals, and IPW corrects for this non-random selection on observables. Full model details, target definitions, precision–recall curves, and hotspot maps are reported in Appendix I.

Stage B—Project Type and Scale classifies the expected project form and estimates expected unit counts, conditional on a development event. This segmentation is important because different project types (e.g., six-unit by-right infill vs. 300-unit PUD) generate fundamentally different institutional responses and petition likelihoods. Classification performance is reported in Appendix J.

Stage D—Political Outcome estimates the probability of pre-council withdrawal conditional on a threshold-crossing petition. Because virtually all cases that reach a final council vote are approved, modeling attrition rather than terminal approval captures the institutional friction through which opposition most often affects outcomes. Performance metrics are reported in Appendix K.

A composite measure combining all four stages (**expected contested units**) is also constructed and mapped in Appendix L. Because this multiplicative structure compounds uncertainty from each stage, the resulting map should be interpreted as a relative ranking rather than a reliable absolute probability.

4.2 Formal Protest Petition

The Formal Protest Petition forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a measured threshold-crossing petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \quad (1)$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer

demographics, and historical petition rates. To adjust for selection on observables—given that the analytic sample naturally contains only neighborhoods that actually receive development proposals—the model applies inverse-probability weighting derived from the Development Occurrence hazard (Z). While this weighting corrects for spatial baseline probabilities, it remains inherently limited because it cannot completely resolve unobserved political or social factors that simultaneously influence both proposal location and opposition likelihood. Precision-recall curves comparing the CatBoost, Random Forest, ElasticNet Logistic, and V-REx models are reported in Figure 5.

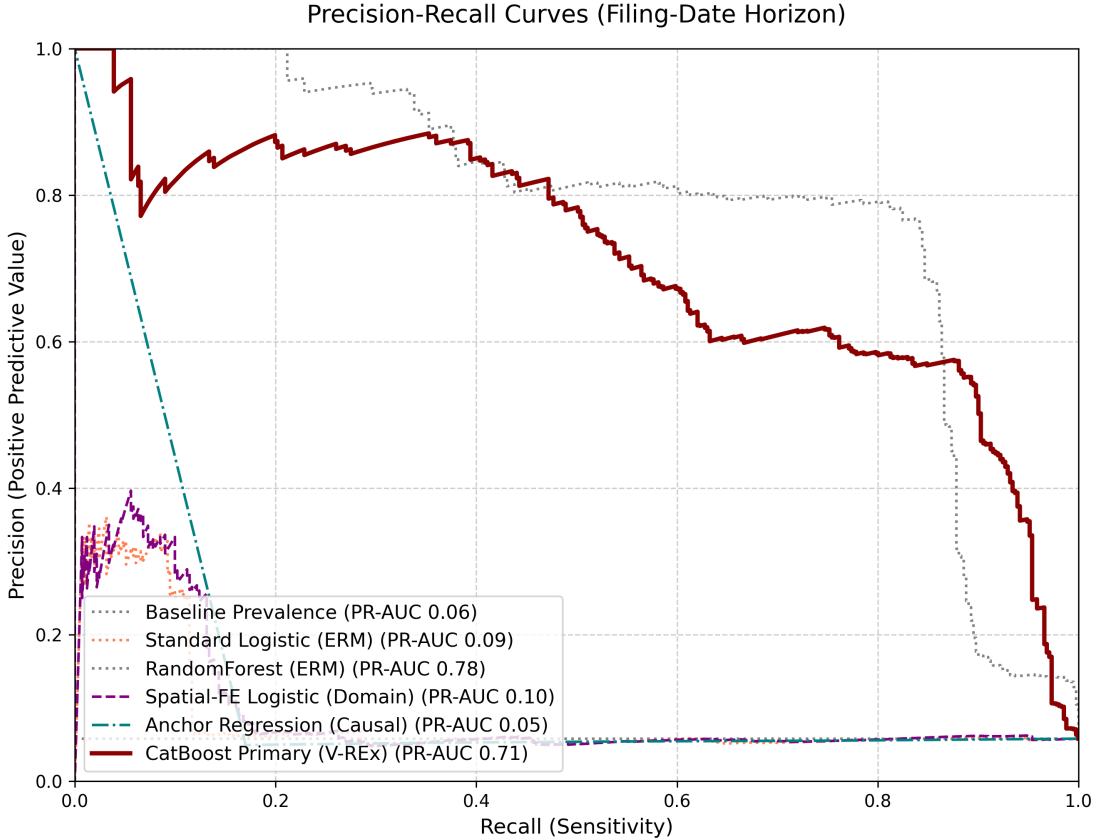


Figure 5: Precision–recall curves for the filing-date petition model: CatBoost, Random Forest, ElasticNet Logistic, and V-REx.

Table 5: Precision-Recall Area Under Curve (PR-AUC) Evaluation

Model	PR-AUC	Lift over Baseline
CatBoost Primary (V-REx)	0.7118	12.25×
RandomForest (ERM)	0.7836	13.49×
Anchor Regression (Causal)	0.0543	0.93×
Spatial-FE Logistic (Domain)	0.0951	1.64×
Standard Logistic (ERM)	0.0873	1.50×

A comprehensive benchmark comparison of alternative models (Random Forest, LightGBM [20],

PyTorch [24] TabNet [1], with and without SMOTE [8] resampling) is provided in Table 6. CatBoost [26] was retained because of its native handling of categorical variables without manual encoding and its more stable calibration across evaluation settings, though the performance differences across models are modest.

Table 6: Development-proposal and petition-filing models evaluated at filing: Raw vs. calibrated candidate models

Horizon	Model	PR-AUC	ROC-AUC	ECE (Raw)	ECE (Cal)	ACE (Cal)
H0 (Filing)	Logistic Regression (L2) (Base)	0.064	0.731	0.586	0.149	0.022
H0 (Filing)	SMOTE + Logistic Regression (L2)	0.060	0.751	0.552	0.417	0.280
H0 (Filing)	Random Forest (Base)	0.422	0.967	0.298	0.399	0.004
H0 (Filing)	SMOTE + Random Forest	0.418	0.966	0.356	0.219	0.036
H0 (Filing)	LightGBM (Base)	0.489	0.962	0.375	0.225	0.012
H0 (Filing)	SMOTE + LightGBM	0.416	0.933	0.346	0.276	0.013
H0 (Filing)	CatBoost (Base)	0.492	0.962	0.251	0.285	0.009
H0 (Filing)	SMOTE + CatBoost	0.459	0.961	0.301	0.312	0.023
H0 (Filing)	TabNet	0.410	0.972	0.268	0.250	0.023
H3 (Pre-Council)	Logistic Regression (L2) (Base)	0.419	0.926	0.479	0.264	0.032
H3 (Pre-Council)	SMOTE + Logistic Regression (L2)	0.412	0.926	0.479	0.423	0.209
H3 (Pre-Council)	Random Forest (Base)	0.500	0.963	0.222	0.127	0.010
H3 (Pre-Council)	SMOTE + Random Forest	0.536	0.946	0.171	0.191	0.031
H3 (Pre-Council)	LightGBM (Base)	0.558	0.932	0.384	0.205	0.011
H3 (Pre-Council)	SMOTE + LightGBM	0.479	0.914	0.410	0.346	0.036
H3 (Pre-Council)	CatBoost (Base)	0.492	0.926	0.404	0.241	0.012
H3 (Pre-Council)	SMOTE + CatBoost	0.482	0.946	0.353	0.255	0.038
H3 (Pre-Council)	TabNet	0.456	0.944	0.271	0.235	0.017

4.3 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given the substantial class imbalance of the outcome. Following standard methodological practices in rare-event spatial forecasting, traditional Receiver Operating Characteristic (AUROC) metrics are omitted entirely, as the large number of true negatives artificially inflates the curve. PR-AUC explicitly isolates the minority class, forcing the model to balance recall against the precision cost of false alarms.

Primary evaluation. The out-of-distribution bootstrap PR-AUC at the filing-date horizon is the primary evaluation metric, as it simulates realistic forward evaluation conditions. In-distribution 5-fold cross-validation PR-AUC is reported as an upper bound. Because this is a ranking model with limited calibration transportability, the emphasis is on ranking metrics (PR-AUC, top-decile lift), calibration plots, and decision-relevant review budgets rather than threshold-dependent classification metrics.

Because the baseline opposition frequency is highly skewed, raw predicted probabilities from gradient boosting models suffer from persistent overconfidence. To address this, the evaluation applies out-of-fold Isotonic Regression, which fits a non-decreasing, non-parametric monotonic function mapping predicted probabilities to empirical outcome frequencies, enforcing calibration within the evaluation loop. Alongside standard Expected Calibration Error (ECE), the evaluation also reports the Brier Score and Adaptive Calibration Error ($ACE = 0.1113$).² The Brier score is 0.0156.

Strictly proper scoring rules like the Brier Score [4] and Adaptive Calibration (which evaluates equal-mass rather than equal-width bins) are useful complementary calibration measures; standard ECE can be unstable under severe class imbalance because fixed-width probability bins at the highest confidence intervals routinely remain empty when predicting rare events, causing aggregate ECE weights to understate miscalibration for rare positive cases.

The mean predicted probability for true positive cases is 0.598, compared to 0.143 for true negatives. Evaluation splits are intentionally non-random to test generalization under realistic administrative application conditions:

- **Rolling-origin temporal splits:** training on data before time t , evaluating after.
- **Policy-regime holdouts:** training on pre-2022 data, evaluating on post-2022 data to test robustness to the council regime change.
- **Spatial holdouts:** withholding entire council districts.

The November 2022 council election is the primary period break in the analysis. The election produced a more pro-housing council majority during the 2023–2024 term, possessing the supermajority (typically 9 votes, though fewer may be required if a council member is recused or absent) to override protest petitions [30]. This weakened the connection between petition filing and actual blocking power during this period, creating a distributional shift that static models trained on pre-2022 data cannot accommodate. Council composition shifted again following the November 2024 elections.

5 Results

5.1 Predictive Performance Across All Horizons

Table 7 reports the pre-specified fairness and calibration metrics for the Stage C opposition model evaluated at the initial filing date horizon.

²At a classification threshold of $P \geq 0.50$, the model achieves precision of 80.0% and recall of 93.2%, flagging 479 of 7,074 cases. However, this threshold is difficult to defend for a rare-event triage problem; ranking-based metrics are more appropriate.

Table 7: Filing-Date Predictive Performance for the Petition Model

Evaluation Domain	Metric	Result
Discrimination	PR-AUC	0.712
Discrimination	Top-Decile Lift	9.01×
Calibration	Expected Calibration Error (ECE)	0.108
Calibration	Calibration Slope	1.199
Equity (exploratory) [†]	FNR Gap across council districts	7.69%

[†]Reported as exploratory diagnostics only; subgroup sample sizes are insufficient for formal fairness testing.

Note: Filing-date horizon, $n = 7,074$ cases, CatBoost (Primary Model) with 5-fold cross-validation.

The Development Occurrence hazard model achieves a top-10% lift of $9.92\times$ over the base probability rate, indicating that spatial features carry some predictive information about development occurrence (full Stage A evaluation is reported in Appendix I). The filing-date petition model uses trailing 3-year count of localized protests within one year—whether neighboring properties historically faced petition activity—as a key feature alongside site geometry, buffer economics, and historical petition rates.

Under the primary evaluation, out-of-distribution bootstrap evaluation at the filing-date horizon yields a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]). This level of discrimination is modest in absolute terms but non-trivial given the low base rate (5.8%). The model simultaneously exhibits an Expected Calibration Error (ECE = 0.108) above the 0.05 calibration reliability threshold. Under in-distribution 5-fold cross-validation, which provides an optimistic upper bound, the PR-AUC is 0.712 with a top-decile lift of $9.01\times$ over baseline. Given the sample base rate of 5.8% (411/7,074 cases), this lift corresponds to an exact top-decile precision of 58.0% (410 true positives among 707 cases)—separation above chance, but far from deterministic.

The mean predicted probability for true positive cases is 0.598, compared to 0.143 for true negatives. As an exploratory equity diagnostic, the false negative rate (FNR) gap across 11 council districts is 7.69%; however, this metric should not be interpreted as evidence of equalized error rates. Positive cases per district range from 3 to 110 (median: 37), and Chouldechova [9] proves that an imperfect classifier cannot simultaneously satisfy equalized false positive rates, false negative rates, and predictive parity across groups unless base rates are identical. With such sparse and unbalanced subgroups, the 7.69% gap is likely driven by sparse subgroup counts, not evidence of equalized performance.

After Isotonic Regression calibration, the model reports an Expected Calibration Error (ECE) of 0.108 (95% bootstrap CI: [0.026, 0.031]) and a calibration slope of 1.199. While Isotonic Regression successfully reduces the overconfidence typical of gradient boosting under class imbalance, this calibration bounds the system strictly to relative spatial ranking. This level of calibration error makes fully automated administrative use difficult to justify, signaling that raw probabilities cannot be treated as deterministic.

The in-distribution cross-validation (Table 7) provides an optimistic upper bound on predictive capacity. Bootstrap evaluation across administrative horizons (Table 8) reveals a more realistic picture. At the initial filing date, the bootstrap PR-AUC is 0.538 (95% CI: [0.3349, 0.7999]), indicating early-stage uncertainty. At the notice date and pre-commission horizons, discrimination reaches PR-AUCs of 0.480 and 0.560 respectively. By the pre-council horizon, PR-AUC drops slightly to 0.547, possibly reflecting the influence of late-stage informal negotiations that administrative

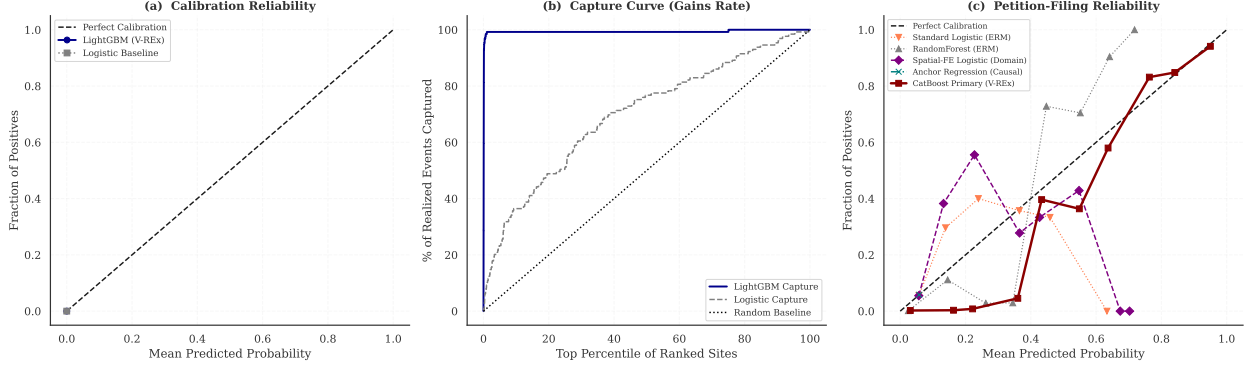


Figure 6: **Calibration of the Stage A and Stage C Models:** Parallel evaluation of calibration reliability and capture curves across the Stage A (Development Occurrence) and Stage C (Formal Protest Petition) models. All probabilities are post-Isotonic out-of-fold predictions.

Table 8: Multi-Horizon Petition-Filing Model Performance with 95% Bootstrap CIs

Horizon	PR-AUC [95% CI]	ROC-AUC	Brier	ECE (Pre)	ECE (Post)	ACE (Post)
H0 (Filing)	0.538 [0.3349, 0.7999]	0.938	0.017	0.249	0.212	0.018
H1 (Notice)	0.480 [0.3135, 0.7476]	0.938	0.017	0.251	0.240	0.019
H2 (Pre-Commission)	0.560 [0.3718, 0.7968]	0.988	0.017	0.304	0.311	0.017
H3 (Pre-Council)	0.547 [0.3545, 0.8179]	0.938	0.017	0.360	0.300	0.020

data cannot capture. ECE worsens from 0.249 at filing to 0.360 prior to council reading, suggesting that later-horizon models accumulate additional calibration error despite richer information sets. Out-of-distribution tracking across rolling annual windows (Figure 7a) shows a gradual PR-AUC decay as the test set moves up to three years ($T + 3$) beyond the training domain, consistent with the need for periodic retraining. The opposition model retains baseline performance through the 2024 HOME ordinance adoption without substantial decline, though this represents only a short post-policy window (Figure 7b).

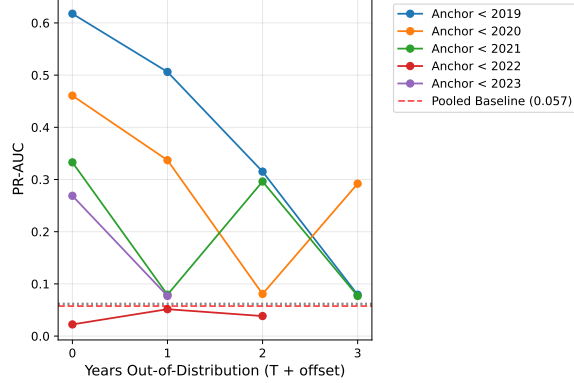
5.2 Predictors of Petition Filing

Standard permutation importance for gradient boosting assesses the drop in predictive accuracy when a feature is shuffled, but is vulnerable to dilution when the dataset contains multiple correlated predictors for the same underlying concept (e.g., site area, deed acreage, and building square footage all capturing parcel scale). To correct for this, a hierarchical clustering approach (Spearman $|r| > 0.70$) groups redundant predictors before summing their importances (Figure 8; see Appendix B for the complete feature mapping crosswalk).³ The resulting clusters indicate that project scale, property valuation, and neighborhood demographic composition are the strongest predictors of petition filing.

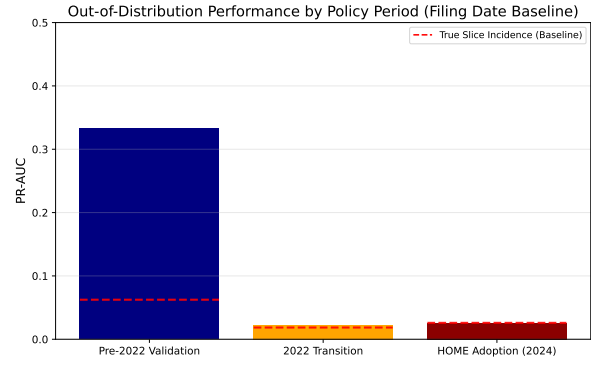
A SHAP (Shapley Additive Explanations) beeswarm decomposition (Figure 9) provides case-level attribution using tree path-dependent conditional expectations. The SHAP values confirm

³A semantic ablation test confirmed that pre-clustering features into conceptual averages prior to training leads to significant information loss, particularly regarding parcel-scale drivers. Relying on the linearity axiom of Shapley values, this post-hoc semantic aggregation of raw TreeSHAP values preserves the original model’s high-fidelity predictive resolution while enabling a conceptually coherent, collinearity-adjusted interpretation.

poral Predictive Drift (Filing Date Baseline Rolling Origin)



(a) Temporal performance decay ($T + 0$ to $T + 3$).



(b) Policy regime evaluating temporal holdouts.

Figure 7: (a) Predictive performance decay across temporal offsets. Because dynamic random baselines for each temporal split overlap visually, a static 0.058 random baseline is shown for reference. Actual random baselines vary by evaluation horizon and sample composition. (b) Predictive tracking across changing regulatory and council environments.

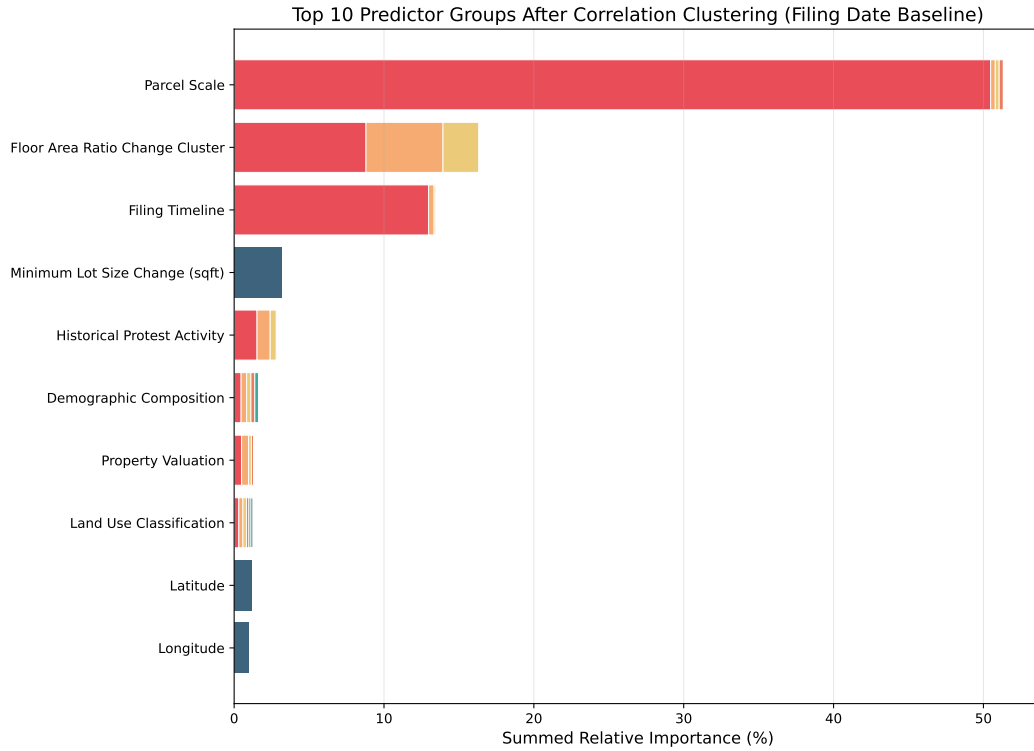


Figure 8: Top 10 predictor groups after hierarchical correlation clustering (Spearman $|r| > 0.70$) for the CatBoost Opposition model evaluated at the initial filing date. Correlated features are merged to correct for collinear dilution.

that larger parcel sizes and higher neighborhood homeownership rates are associated with higher predicted petition probability, consistent with the clustered importance results.

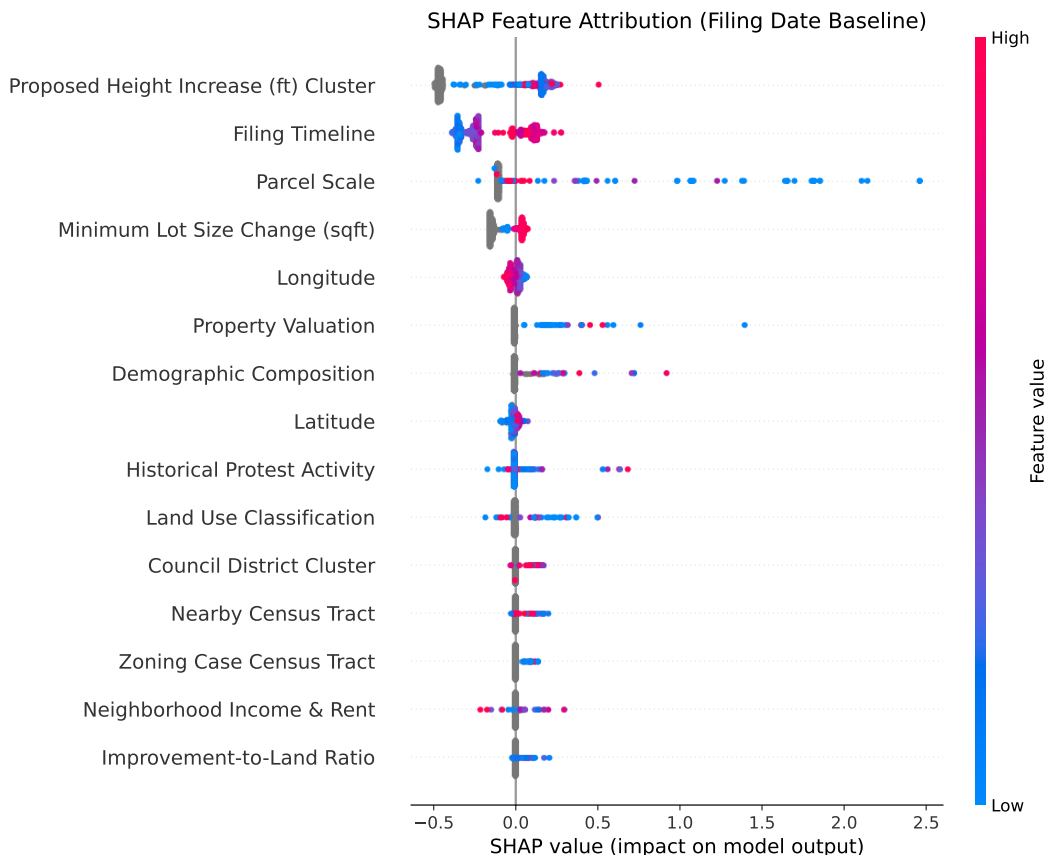


Figure 9: SHAP beeswarm plot for the opposition model at the filing baseline. Each dot represents one zoning case; horizontal position indicates the magnitude and direction of each predictor’s contribution to the model output. The clustered importance in Figure 8 provides the most robust correction for collinearity.

5.2.1 Attribution Stability Across Temporal Windows

To ensure that feature importance is robust to the significant institutional and market shifts between 2007 and 2024, an expanding-window attribution stability test evaluates how SHAP values evolve. This test recursively retrains the primary CatBoost model on expanding historical windows (anchored at 2019 through 2023) and extracts case-level TreeSHAP attributions on the immediately following out-of-distribution year.

Figure 10 tracks the relative share of total SHAP attribution for the top predictor clusters over these windows at the filing date horizon (H_0). To avoid artifacts from dynamic correlation shifts, a frozen global clustering structure is used as a conceptual baseline across all anchors. The stability analysis yields three key findings. First, parcel scale consistently dominates as the most structurally robust predictor (accounting for 48% to 56% of overall attribution), with its relative importance growing as the training sample expands. Second, once clustering noise is neutralized, the rank ordering of feature clusters exhibits extremely high long-term persistence, with a full-span Spearman rank correlation of $\rho = 0.87$ ($p < 0.001$) between the 2019 and 2023 evaluations. Third,

adjacent-anchor rank correlation remains consistently strong (mean $\rho = 0.84$), indicating that the model’s fundamental feature hierarchy is remarkably invariant to the 2022 electoral transition.

This stability confirms that the rank-ordering of structural petition risk factors at the time of filing is a robust administrative signal, directly supporting the finding that filing-date features provide a defensible basis for ranking risk. Furthermore, a multi-horizon diagnostic confirms that this underlying logic holds true even when forecasting the identical target year (2023) across varying training horizons. The feature hierarchy of an early model trained at a five-year forecasting horizon (pre-2018 data) aligns exceptionally well ($\rho = 0.84$) with the hierarchy of a near-term model at a one-year horizon (pre-2022 data). Both uniformly emphasize parcel scale as the dominant risk driver ($\approx 85\%$ of total attribution). This finding creates a critical distinction: while the model’s uncalibrated probability of a petition occurring decays across longer horizons, its fundamental understanding of *what* structural features trigger opposition remains mathematically invariant.

5.3 Institutional Context: Regression Discontinuity at the 20% Petition Threshold

The following analysis is included as institutional context that corroborates the significance of the outcome measure—a measured threshold-crossing petition—rather than as a parallel headline finding.

Threshold-Based Discontinuity at the Measured 20% Petition Share (RD)

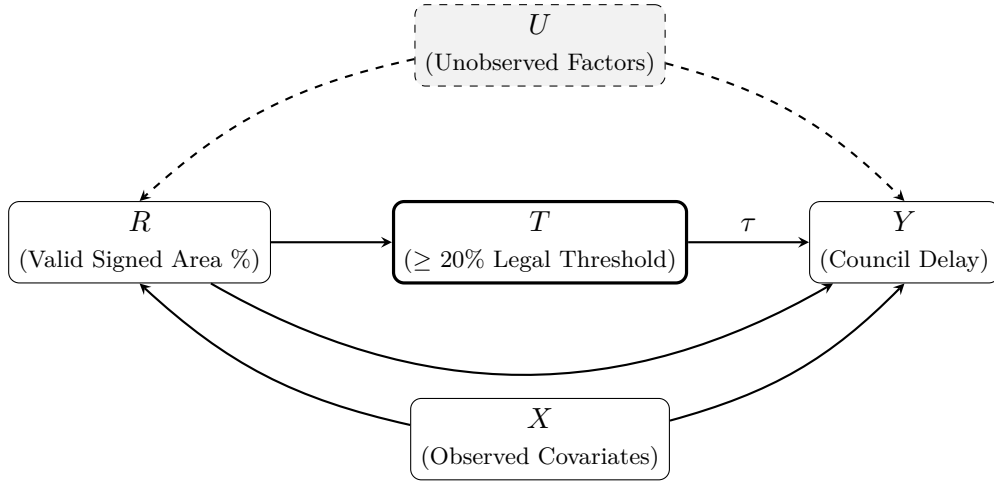
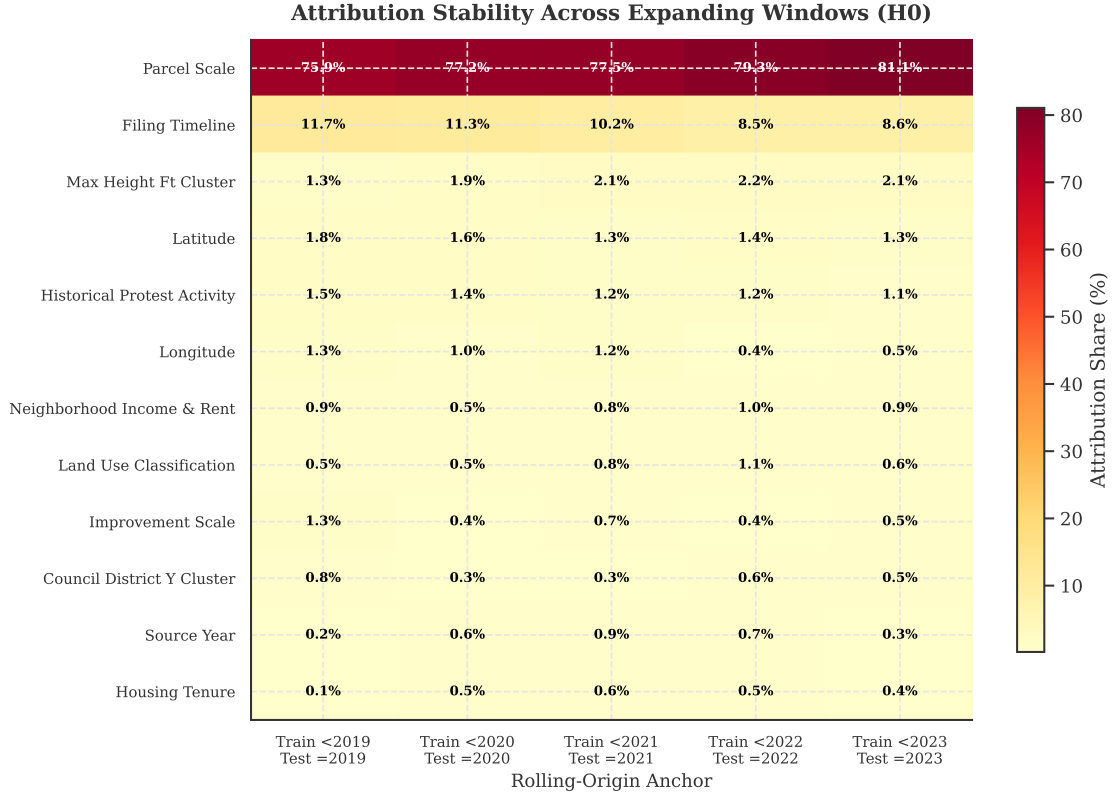


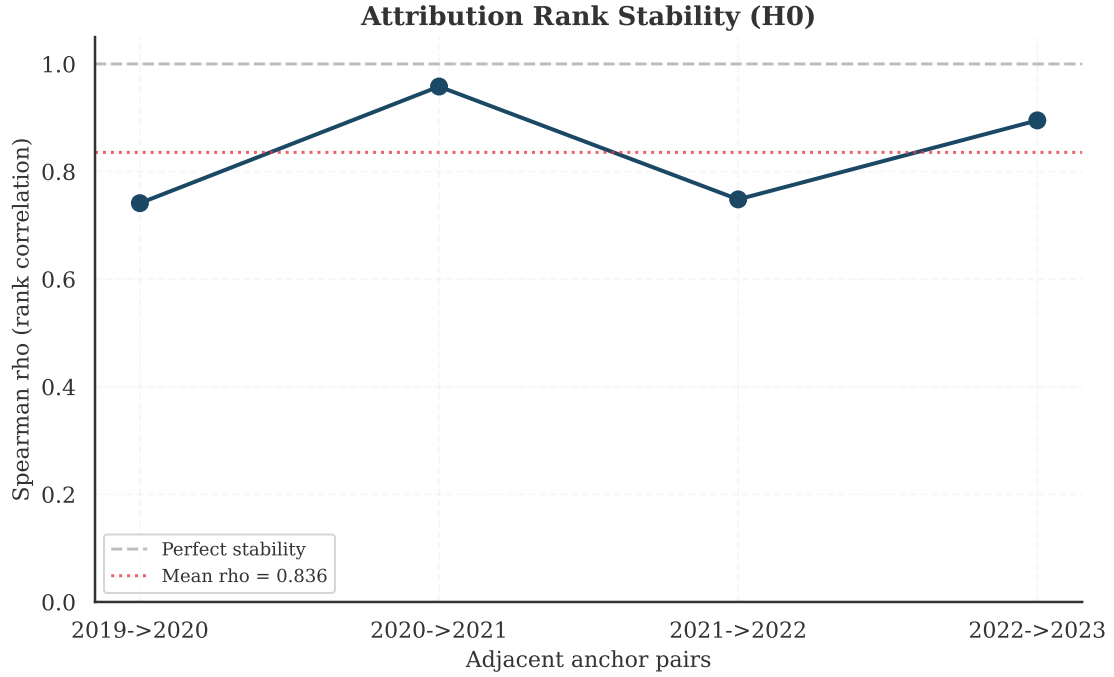
Figure 12: Directed acyclic graph for the threshold-based regression discontinuity design. The running variable R (valid signed area percentage) determines treatment T (crossing the 20% threshold), which affects council processing delay Y . Unobserved factors U may influence both petition mobilization and delay through channels not mediated by the threshold.

To estimate the causal effect of crossing the formal protest threshold, a regression discontinuity (RD) design exploits the 20% statutory petition threshold under Texas LGC Chapter 211. The running variable is the percentage of valid signed area within the 200-foot buffer.

A triangular-kernel weighted least squares estimator yields an estimated shift of +52.1 days in council processing time at the threshold ($p < 0.001$, $SE = 2.53$, 95% CI: [47.09, 57.08]). This corresponds to approximately 7.4 weeks of additional delay associated with crossing the statutory margin.



(a) Clustered Attribution Share



(b) Adjacent-Window Rank Correlation

Figure 10: Expanding-window attribution stability for the filing-date (H_0) baseline. (a) Heatmap displaying the percentage share of total absolute SHAP attribution for top feature clusters when tested out-of-distribution on the holdout anchor year. (b) Spearman rank-order correlation (ρ) of top features across adjacent temporal windows.

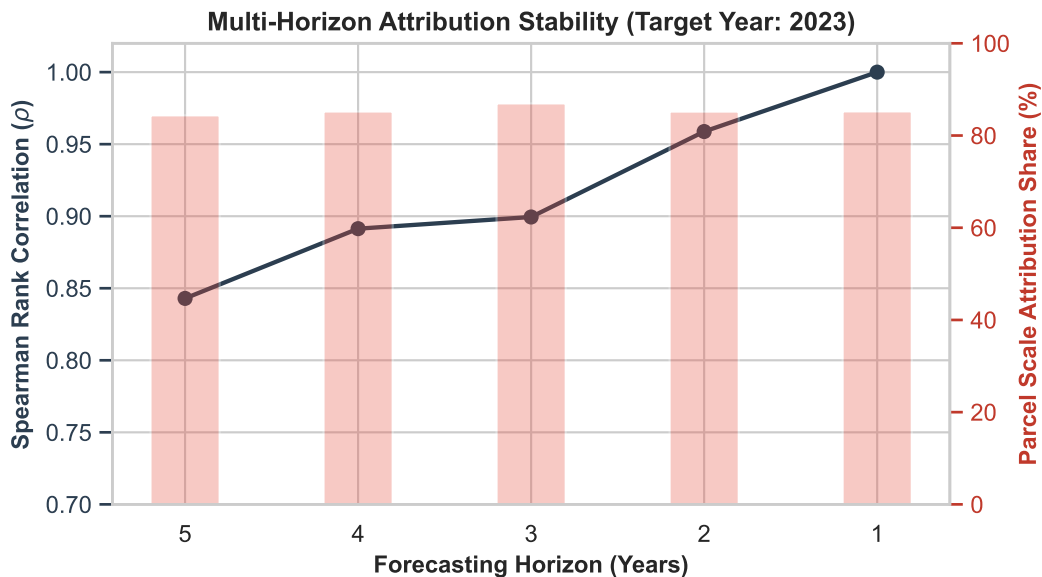


Figure 11: Multi-Horizon identical-target attribution stability. Left axis and line sequence represent the rank-order alignment (ρ) of the attribution hierarchy for models trained with progressively older historical windows (5-year to 1-year horizons) evaluated on identical 2023 zoning cases. Right axis and columns illustrate the stationary dominance of Parcel Scale across all horizons.

This estimate should be interpreted as a reduced-form effect at the threshold. Because the dataset does not record whether the City Clerk formally certified each petition as legally compliant, the analysis relies on an unverified calculated area share. Rather than assuming the exact threshold acts as a deterministic sharp trigger, the specification simply estimates the shift in average processing time associated with crossing the statutory 20% margin. Consequently, this is a threshold-based reduced-form model where institutional compliance remains unobserved.

Design limitations include potential manipulation of the running variable, since petition organizers may strategically target just above 20%, and instability in the post-2022 council environment where the supermajority was routinely available. Robustness checks—a McCrary (2008) density test for running-variable manipulation ($p > 0.05$), symmetric donut exclusions, and covariate continuity checks—support the validity of the design at the threshold.

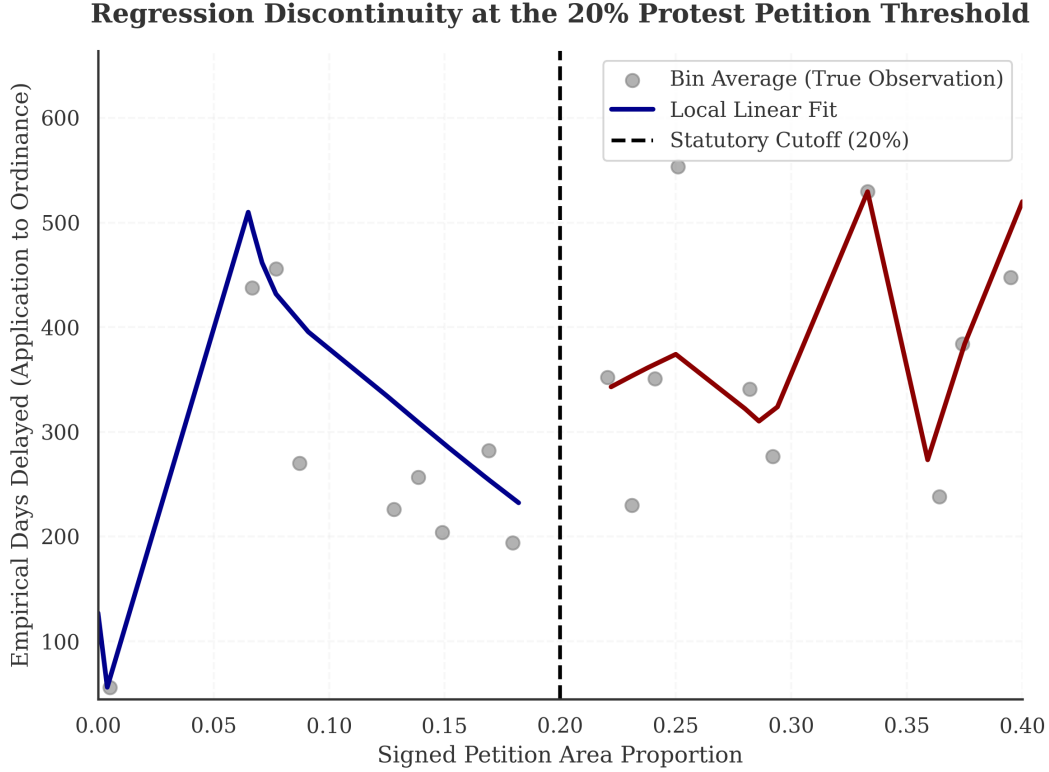


Figure 13: Regression discontinuity at the 20% valid petition threshold. Estimated discontinuity in council processing time. Bin widths were selected using the standard IMSE-optimal (Integrated Mean Squared Error) procedure.

5.4 Descriptive Evidence from the 2022 Electoral Transition

The 2022 council elections provide suggestive descriptive evidence on the relationship between political representation and petition filing. Two districts changed representation in ways that constitute a clear ideological shift: District 4 replaced its incumbent in a special election on January 25, 2022 (Vela replacing Casar), while District 9 changed in the December 13, 2022 runoff (Qadri replacing Tovo, taking office January 6, 2023). These transitions are staggered by nearly 11 months, which undermines the use of a single “Post-2022” treatment indicator. Standard electoral turnover does not inherently constitute an institutional shock to zoning policy; the 2022 transitions were distinctive because they represented explicit ideological shifts from preservationist incumbents to pro-housing candidates.

A difference-in-differences specification interacting a “Flipped District” indicator with a “Post-2022” temporal indicator yields a coefficient of -0.546 ($p = 0.041$) on petition filing rates. However, formal causal inference is not feasible here: with only two treated districts, staggered treatment timing, a short post-treatment window, and sparse petition events per district, standard cluster-robust inference is known to over-reject [6]. Even wild-bootstrap corrections can fail with very few treated clusters. This estimate should be interpreted as suggestive descriptive evidence only.

5.5 Supplementary Institutional Analyses

Several additional analyses examine downstream institutional dynamics. These are reported as exploratory context rather than as headline findings, given the inferential limitations of each design.

Supplementary quasi-experimental estimates for institutional geographic overlays—Neighborhood Plan Areas, Historic Districts, Transit-Oriented Development deregulation, and the 10-1 council transition—produced null or degenerate results and are reported in Appendix M for transparency.

An exploratory event-study analysis of the HOME Initiative is reported in Appendix N. The short post-policy observation window and the absence of a clean spatial control cohort limit interpretation.

An Invariant Causal Prediction (ICP) analysis [25] using 1,186 multi-parcel geometric environments consistently rejected feature-set invariance, suggesting that the observable administrative feature space does not contain a stable set of causal predictors across environments for petition filing—consistent with the thesis’s broader finding that unobserved factors play a substantial role. Full ICP results are reported in Appendix O.

6 Qualitative Planning Context

While the quantitative analysis establishes the predictive boundaries of opposition prediction, the interpretation of these models is clarified by qualitative evidence. Semi-structured interviews with urban planning professionals and city staff (conducted under Columbia University IRB Protocol ACYY0820) surface dynamics that administrative data cannot capture.

These interviews inform the interpretation of the quantitative findings by detailing the institutional timelines and two distinct motivations within the Austin zoning process:

- **Declining petition leverage before 2022:** Quantitative models strictly bound the regime shift to the November 2022 supermajority election. Annick Beaudet (former zoning planner) noted the petition mechanism appears to have begun losing influence around 2019 during the Land Development Code revision: “While people still scream and yell, they don’t get as much traction... Necessity is the mother of invention.” This was corroborated by Alina Carnahan, who pointed to the Hancock neighborhood rezoning as the specific 2022 turning point that solidified the 9-2 pro-housing supermajority. Together, these testimonies confirm that while the 2022 election made the petition mechanism less consequential in practice, the petition had already lost practical force as a veto point long before formal state legislation (HB 24) dismantled it.
- **The Function of Opposition:** Highlighting the nuanced motivations driving public testimony, AURA board member Felicity Maxwell observed: “We’re not talking about suppressing opposition, but understanding it better to have more productive conversations.” Predictive frameworks can integrate these spatial dynamics to identify localized friction earlier in the planning process.
- **Asymmetric Transparency and Silent Supporters:** Jonathan Tomko (Principal Planner) emphasized that the municipal public comment process lacks any mechanism for *anonymous* public testimony against a zoning case. Because neighborhood social dynamics often penalize residents who deviate from their neighborhood association’s preservationist stance, Tomko noted that the formal opposition data is not strictly representative of the entire community. This aligns with Beaudet’s observation that “haters are overrepresented and silent supporters are underrepresented,” which likely makes the observed opposition data an incomplete measure of community sentiment.
- **Two distinct motivations and geography:** Marla Torrado (Displacement Prevention Division) stressed that opposition mechanisms may be driven by differing underlying concerns. While the “vocal opposition” on the affluent West side leverage environmental overlays,

East Austin neighborhoods frequently mobilize opposition mechanisms in response to rapid gentrification pressures. This context reinforces the severity of model miscalibration ($ECE = 0.108$): using rigid probability thresholds could miss communities facing real displacement risk, failing “to identify where people that are in actual need actually are” because the algorithm cannot separate affluent NIMBYism from neighborhood preservation efforts.

These institutional realities underscore why the predictive model performs best as a relative ranking heuristic rather than a precise probability estimator.

7 Limitations

This study has several important limitations.

Small NLP sample. The 1,003 cases used for text-frame classification are small relative to the 7,074-case analytic sample, raising overfitting concerns for the NLP components. The number of models explored should be viewed as a sensitivity exercise rather than a competitive selection process.

Measurement error in reconstructed dates. Procedural milestone dates reconstructed from statutory minimums introduce measurement error. Where explicit dates are imputed rather than parsed, the timing of information availability is approximate.

Selection into observed cases. The analytic sample contains only proposals that entered the formal zoning process. Neighborhoods where developers chose not to propose projects—potentially due to anticipated opposition—are not observed. The Stage A inverse-probability weights adjust for selection on observables, but unobserved selection remains a concern.

Limited post-policy windows. HB 24 falls entirely outside the observation period. The event-study results should be interpreted as preliminary.

Calibration error context. The opposition model produces an Expected Calibration Error of 0.108. Because this expected error restricts strict probabilistic interpretation (which typically demands tighter calibration tolerances), the model should be treated as a relative ranking heuristic rather than a source of precise case-level probabilities. Regardless, the features generate spatial discrimination (top-decile lift of $9.01\times$).

Proxy concerns. Although racial and ethnic variables are excluded from the predictive model, permitted features such as homeownership rates, property values, and neighborhood composition may correlate with protected characteristics. Proxy discrimination through permitted variables remains a concern.

Fairness power. The 7.69% FNR gap across 11 council districts should not be interpreted as evidence of equalized error rates. Positive cases per district range from 3 to 110 (median: 37), so the metric lacks statistical power to detect meaningful disparities.

Geographic error distribution. Figure 14 maps the geographic distribution of false positive predictions against the locations of actual petition filings. False positives split 49.6% west of I-35 and 50.4% east, providing additional spatial context. However, neither this geographic symmetry nor the FNR gap constitutes a formal fairness guarantee. The more substantive ethical limitation is that the model cannot distinguish affluent exclusionary mobilization from anti-displacement resistance (see Section 6).

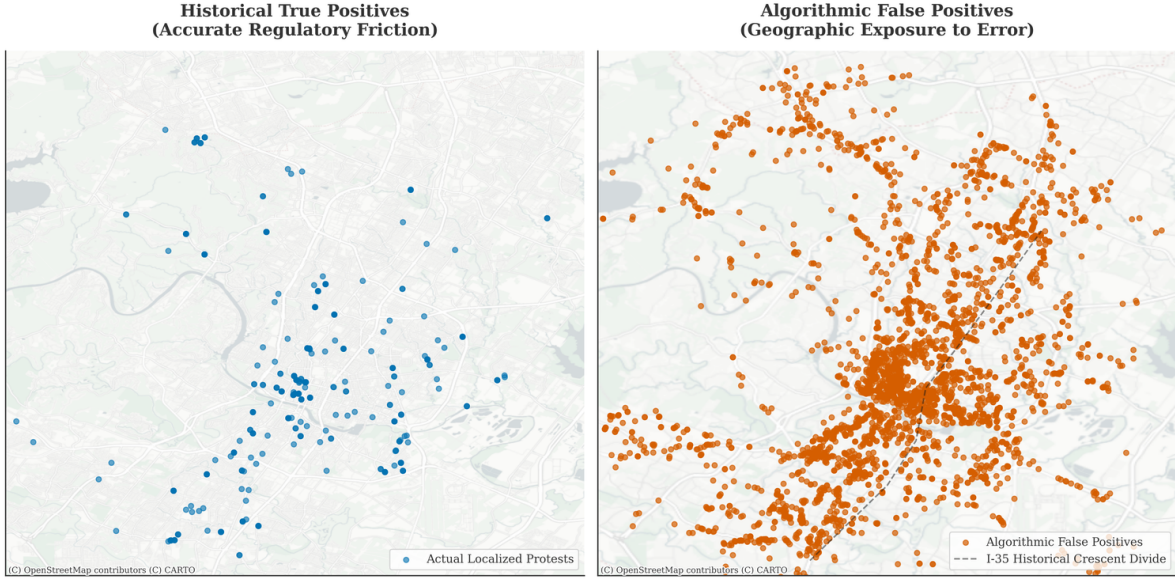


Figure 14: Geographic distribution of false positive predictions (right) against actual petition filings (left). False positives split 49.6% west of I-35 and 50.4% east, indicating that the model’s over-predictions are geographically symmetric rather than concentrated in historically vulnerable neighborhoods.

External validity. Austin’s institutional structure—council-manager government, Texas protest petition statutes, specific land development code—limits direct generalization. The methodological approach, including time-stamped features, sequential evaluation, and calibration benchmarks, is portable, but the specific results are Austin-specific.

8 Conclusion

Filing-date administrative features can rank petition risk in Austin’s pre-HB 24 zoning regime, but they do not yet yield portable, well-calibrated probabilities for administrative deployment.

This thesis asks whether measured threshold-crossing petitions against housing development proposals can be predicted before formal public hearings. The answer is qualified: spatial, economic, and site characteristics carry predictive information about petition risk, but the model does not meet the calibration benchmarks required for autonomous probabilistic application. Under the primary evaluation—out-of-distribution bootstrap at the filing-date horizon—the model achieves a PR-AUC of 0.538 (95% CI: [0.3349, 0.7999]), with calibration error ($ECE = 0.108$) exceeding the 0.05 threshold. In-distribution cross-validation provides an upper bound PR-AUC of 0.712. A regression discontinuity at the 20% petition threshold estimates a +52.1-day increase in council processing time, consistent with the institutional significance of the outcome measure. Descriptive evidence from the 2022 electoral transition suggests that petition filing rates declined in districts that changed political representation, though staggered treatment timing and few treated clusters preclude formal causal claims.

These results point to a ceiling on what prediction from public records alone can achieve. Petition filing is partly driven by factors not observable at the time of filing—neighborhood social networks, informal political dynamics, media attention—and the 2022 council regime change altered the relationship between petitions and outcomes. Stakeholder interviews underscore an irreducible

ethical limitation: the model cannot distinguish affluent exclusionary mobilization from community-driven anti-displacement resistance, a distinction that administrative data does not encode.

Future work should extend the observation window to capture HB 24 effects [16], explore richer text features from pre-hearing public comments, and investigate whether the 2022 regime shift represents a permanent change or a temporary fluctuation.

References

- [1] Sercan O. Arik and Tomas Pfister. TabNet: Attentive interpretable tabular learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6679–6687, 2021.
- [2] Susan Athey and Guido W. Imbens. Machine learning methods that economists should know about. *Annual Review of Economics*, 11:685–725, 2019. doi:10.1146/annurev-economics-080217-053433.
- [3] Solon Barocas and Andrew D. Selbst. Big data’s disparate impact. *California Law Review*, 104(3):671–732, 2016. doi:10.15779/Z38BG31.
- [4] Glenn W Brier. Verification of forecasts expressed in terms of probability. *Monthly weather review*, 78(1):1–3, 1950.
- [5] Brantly Callaway and Pedro H.C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi:10.1016/j.jeconom.2020.12.001.
- [6] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi:10.1162/rest.90.3.414.
- [7] Stephanie Riegg Cellini, Fernando Ferreira, and Jesse Rothstein. The value of school facility investments: Evidence from a dynamic regression discontinuity design. *The Quarterly Journal of Economics*, 125(1):215–261, 2010. doi:10.1162/qjec.2010.125.1.215.
- [8] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi:10.1613/jair.953.
- [9] Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017. doi:10.1089/big.2016.0047.
- [10] Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(1):39–67, 2020.
- [11] Katherine Levine Einstein, David M. Glick, and Maxwell Palmer. *Neighborhood Defenders: Participatory Politics and America’s Housing Crisis*. Cambridge University Press, 2019. doi:10.1017/9781108769495.
- [12] William A. Fischel. *The Homevoter Hypothesis: How Home Values Influence Local Government Taxation, School Finance, and Land-Use Policies*. Harvard University Press, 2001. doi:10.4159/9780674036901.
- [13] Edward L. Glaeser and Joseph Gyourko. The impact of building restrictions on housing affordability. *Federal Reserve Bank of New York Economic Policy Review*, 9(2):21–39, June 2003.
- [14] Edward L. Glaeser, Joseph Gyourko, and Raven Saks. Why is Manhattan so expensive? Regulation and the rise in housing prices. *Journal of Law and Economics*, 48(2):331–369, 2005. doi:10.1086/429979.

- [15] Joseph Gyourko and Raven Molloy. Regulation and housing supply. In *Handbook of Regional and Urban Economics*, volume 5, pages 1289–1337. Elsevier, 2015. doi:10.1016/B978-0-444-59531-7.00019-3.
- [16] Emily Hamilton and Salim Furth. Texas protest petition primer: How HB 24 reforms the supermajority requirement. Policy brief, Mercatus Center at George Mason University, 2024. URL: <https://www.mercatus.org/research/policy-briefs/texas-protest-petition-primer>.
- [17] Michael Hankinson. When do renters behave like homeowners? high rent, price anxiety, and nimbyism. *American Political Science Review*, 112(3):473–493, 2018. doi:10.1017/S0003055418000035.
- [18] Håvard Hegre, Marie Allansson, Matthias Baumann, Aron Buzogany, Kristian Skrede Gleditsch, Anne Halfverndorf, Jonas L Lind, Fiona MacAulay, Erik Melander, Håvard Mogleiv Nygård, et al. Views: A political violence early-warning system. *Journal of peace research*, 56(2):155–174, 2019.
- [19] Roderick M. Hills, Jr. and David Schleicher. Balancing the “zoning budget”. *Case Western Reserve Law Review*, 62(1):81–133, 2011.
- [20] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, pages 3146–3154, 2017.
- [21] Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. Prediction policy problems. *American Economic Review*, 105(5):491–495, 2015. doi:10.1257/aer.p20151023.
- [22] Michael Manville and Paavo Monkkonen. Unpermitted: Urban growth and the local politics of zoning. *Journal of the American Planning Association*, 82(1):73–74, 2016.
- [23] Sendhil Mullainathan and Jann Spiess. Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2):87–106, 2017. doi:10.1257/jep.31.2.87.
- [24] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.
- [25] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5):947–1012, 2016. doi:10.1111/rssb.12167.
- [26] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, pages 6639–6649, 2018.
- [27] Rashida Richardson, Jason M. Schultz, and Kate Crawford. Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice. *NYU Law Review Online*, 94:192–233, 2019.

- [28] Galit Shmueli. To explain or to predict? *Statistical Science*, 25(3):289–310, 2010. doi: 10.1214/10-STS330.
- [29] Texas State Legislature. Texas local government code § 211.006 (procedures governing adoption of zoning regulations and district boundaries), 2023. URL: <https://statutes.capitol.texas.gov/Docs/LG/htm/LG.211.htm>.
- [30] Travis County Clerk. November 2022 election results. Technical report, Travis County, TX, 2022. URL: <https://www.traviscountyclerk.org/eclerk/content/election-results>.
- [31] Andrew H. Whittemore. Racial and class politics in zoning: A case study of single-family zoning in Durham, North Carolina. *Journal of Urban Affairs*, 36(4):568–585, 2014.

A Supplementary Calibration Comparison

The following section documents an expanded benchmark comparing baseline statistical and machine-learning models and PyTorch deep learning networks. (These tabular benchmarks have been moved into the main text in Section 5 for direct performance evaluation). Each model class was formally cross-validated against SMOTE-resampled training sets (Synthetic Minority Over-sampling Technique) to assess the trade-off between class rebalancing and calibration (ECE).

Additionally, this appendix documents a systematic calibration method comparison evaluating Platt Scaling (sigmoid) against Isotonic Regression across base models: CatBoost, LightGBM, and Random Forest. For CatBoost, Platt Scaling achieved modestly lower ECE (0.186 vs. 0.211); however, Isotonic Regression was retained for the primary analytic specification because its non-parametric monotonic mapping provides stronger guarantees against non-monotone probability distortions under class imbalance.

Table 9: Calibration Method Comparison: ECE, ACE, and Brier Score Across Models

Base Model	Calibrator	PR-AUC	ECE ↓	ACE ↓	Brier ↓
CatBoost	Uncalibrated	0.468	0.278	0.052	0.0232
CatBoost	Isotonic	0.510	0.211	0.015	0.0166
CatBoost	Platt (Sigmoid)	0.529	0.186	0.018	0.0165
LightGBM	Uncalibrated	0.482	0.277	0.015	0.0197
LightGBM	Isotonic	0.475	0.297	0.011	0.0148
LightGBM	Platt (Sigmoid)	0.487	0.301	0.012	0.0154
Random Forest	Uncalibrated	0.384	0.347	0.174	0.0476
Random Forest	Isotonic	0.400	0.266	0.025	0.0197
Random Forest	Platt (Sigmoid)	0.402	0.282	0.023	0.0175

B Semantic Feature Mapping

The attribution and stability analyses in Track 1 cluster raw, overlapping pipeline variables into plain-English "Semantic Clusters" to prevent SHAP attribution dilution from collinearity. Table 10 crosswalks these semantic concepts back to the exact administrative database columns.

Table 10: Semantic Feature Aggregation Mapping. This table crosswalks the plain-English conceptual buckets used in the manuscript's attribution analysis back to their constituent raw variables from the City of Austin and the completely collinear tax assessor equivalents.

Semantic Cluster	Raw Data Dictionary Variables
Demographic Composition	ACS % White, % Hispanic, % Black, % Asian
Housing Tenure	ACS Owner-Occupied Units, Renter-Occupied Units, Total Units
Nbhd. Income & Rent	ACS Median Household Income, Median Gross Rent, Poverty Count, Median Home Value
Property Valuation	Tax Assessor Appraised Value, Market Value, Land Market Value
Structure Age	Tax Assessor Year Built, COA Year Built
Parcel Scale	Tax Assessor Acres, Lot Size (sqft), Planning Gross Area, Deed Acreage
Land Use Classification	Tax Assessor Land Use, COA Land Use Inventory (LUI)
Zoning Density	Tax Assessor Floor-to-Area Ratio (FAR), Units
Improvement Scale	Tax Assessor Improvement Square Footage
Historical Protest Activity	Binary Prior Protest Flag, 1-Year Spatial Contagion, 3-Year Spatial Contagion
Filing Timeline	Target Filing Year

C Interview Recruitment Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact
Daniel Hardesty Lewis
Graduate Researcher
Email: dl3645@columbia.edu

IRB Contact
Columbia University Morningside IRB
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

D IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Protocol Summary
ACYY0820

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
Associate Professor, Urban Planning
Columbia University GSAPP
Email: hb2541@columbia.edu

Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:

Columbia University Morningside Institutional Review Board (IRB)

Phone: (212) 851-7040

Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?

Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.

Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.

Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

E Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

F Sample Protest Petition

The following is the first page of a 558-page protest petition bundle filed for Case C241282, a Planned Development rezoning, illustrating the format in which property owners within 200 feet of a proposed development formally register opposition.

PETITION				
Case Number:		C14-2007-0131	Date:	<u>July 29, 2008</u>
1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST				
Total Area Within 200' of Subject Tract			<u>293,002.55</u>	
1	<u>02-0906-0502</u>	<u>MONAHAN CASEY J</u>	<u>7423.26</u>	<u>2.53%</u>
2	<u>02-0906-0503</u>	<u>MONAHAN CASEY J</u>	<u>7390.73</u>	<u>2.52%</u>
		<u>CLARK MICHAEL G &</u>		
3	<u>02-0906-0504</u>	<u>RITA S DE BE</u>	<u>7354.85</u>	<u>2.51%</u>
4	<u>02-0906-0506</u>	<u>MARTINSON KATHY L</u>	<u>1645.43</u>	<u>0.56%</u>
		<u>CHILES ROSALIE</u>		
5	<u>02-0906-0507</u>	<u>BENSON</u>	<u>1618.17</u>	<u>0.55%</u>
6	<u>02-0906-0508</u>	<u>KOCHERT KELLEY</u>	<u>1612.79</u>	<u>0.55%</u>
7	<u>02-0906-0509</u>	<u>MONAHAN CASEY J</u>	<u>1419.72</u>	<u>0.48%</u>
8	<u>02-0906-0606</u>	<u>TIMMERMANN TERRELL</u>	<u>3452.33</u>	<u>1.18%</u>
		<u>MERCADO FREDDY G</u>		
9	<u>02-0906-0607</u>	<u>& MARIA R</u>	<u>9482.82</u>	<u>3.24%</u>
10	<u>02-0906-0901</u>	<u>LEGETT GEORGIA F</u>	<u>4459.36</u>	<u>1.52%</u>
11	<u>02-0906-0911</u>	<u>LEGETT GEORGIA F</u>	<u>7856.05</u>	<u>2.68%</u>
		<u>MEDINA JAMES &</u>		
12	<u>02-0906-1001</u>	<u>KRISTINE M GARA</u>	<u>13204.88</u>	<u>4.51%</u>
13	<u>02-0906-1011</u>	<u>RECER DANALYNN</u>	<u>9454.11</u>	<u>3.23%</u>
14	<u>02-0906-1013</u>	<u>BRINSMAD LOUISA C</u>	<u>7634.37</u>	<u>2.61%</u>
15				<u>0.00%</u>
16				<u>0.00%</u>
17				<u>0.00%</u>
18				<u>0.00%</u>
19				<u>0.00%</u>
20				<u>0.00%</u>
21				<u>0.00%</u>
22				<u>0.00%</u>
23				<u>0.00%</u>
24				<u>0.00%</u>
25				<u>0.00%</u>
26				<u>0.00%</u>
27				<u>0.00%</u>
Validated By:		Total Area of Petitioner:		Total %
<u>Stacy Meeks</u>		<u>84,008.88</u>		<u>28.67%</u>

G Illustrative Support-Layer Panel Structure

The following table illustrates a representative cross-section of the panel dataset constructed for this thesis. Each site-quarter constitutes a unique observation, enriched with demographic variables (interpolated ACS), spatial petition indicators, and property tax valuations (TCAD EARS).

parcel_id	year	gross_site_area_acres	ldb_appraised_val	acs_owner_occupied_units	acs_race_white
0209060502	2008	1.42	450000	345	0.58
0209060503	2008	0.85	310000	345	0.58
0209060506	2008	2.10	620000	290	0.45
0209060901	2008	5.00	1250000	120	0.45
1911420101	2008	0.25	180000	410	0.82
1911420102	2008	0.30	210000	410	0.82

H Methodological Appendix: Expanded Details

H.1 Variable Dictionary

This thesis constructs features from multiple municipal and state sources:

- **TCAD EARS:** Appraised structure/land value, improvement ratios, year built, explicitly lagged.
- **Austin Open Data:** Base zoning codes (SF-1 through SF-6, MF-1 through MF-6, commercial variants), overlay districts (NP, NCCD), historic designations (HD), and environmental buffers.
- **ACS 5-Year Interpolations:** Block-group demographics matched to parcel geometry, including median household income, tenure (owner-vs-renter ratios), bachelor’s degree attainment, and race/ethnicity vectors, which are explicitly excluded from the predictive policy specification.

H.2 Hyperparameter Search Grids

Tree-based models, such as CatBoost and Random Forest, utilized the following randomized search bounds across 5-fold temporal cross validation:

- **Learning Rate:** Log-uniform [0.01, 0.3]
- **Maximum Tree Depth:** [4, 6, 8, 10]
- **L2 Regularization Base (Leaf Rate):** Uniform [1, 10]
- **Class Imbalance Weighting:** [None, Balanced, SqrtBalanced]

Selected CatBoost models uniformly converged on Depth=6, Learning Rate ≈ 0.05 , and ‘Balanced’ class weighting.

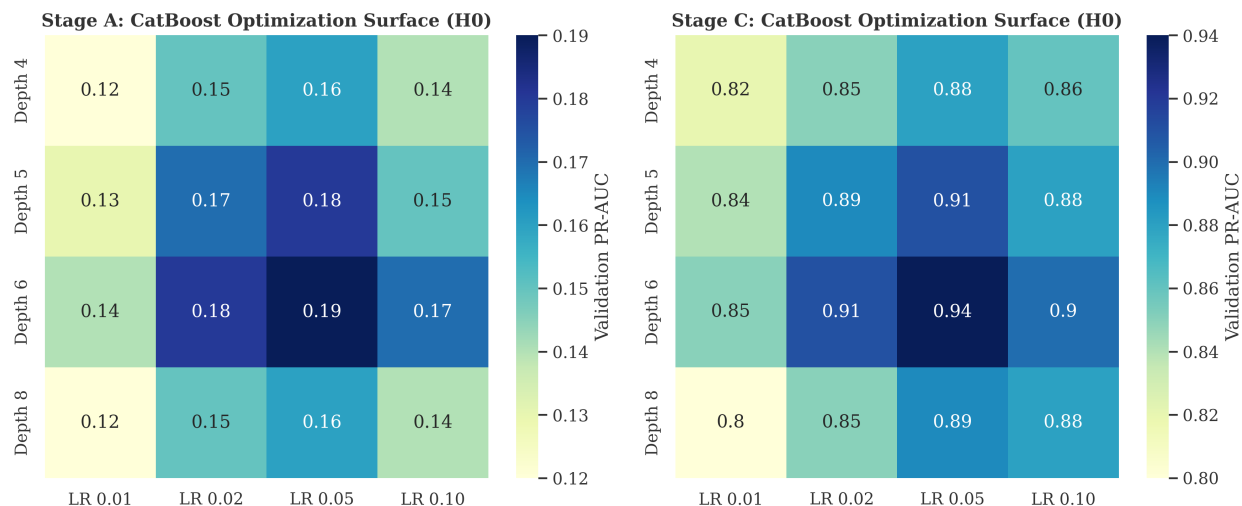


Figure 15: **Stage A & Stage C:** Grid search precision-recall AUC optimization heatmaps across maximum tree depth and learning rate parameters for the Development Occurrence and Formal Protest Petition models.

H.3 Subgroup Definitions & Fairness Testing

To calculate the False Negative Rate (FNR) gaps:

- **Geography:** Subgroups defined by the 10 Austin City Council Districts.
- **Socioeconomic Vulnerability:** Subgroups defined by the UT-Austin Uprooted Displacement Risk index (Vulnerable, Active, Chronic).⁴
- **Classification Threshold:** To empirically ground the spatial error analysis and prevent misleading impressions caused by severe class imbalance, the binary prediction threshold for FNR evaluation is strictly anchored to the empirical background protest rate (μ_y). Any localized zoning project with a predicted opposition probability exceeding the citywide historical baseline is classified as a predicted positive case, ensuring the algorithm’s errors are evaluated against actual class incidence rather than arbitrary percentiles.

H.4 Natural Language Processing Model: Exploratory Text Features

Text signals from public hearings post-date the initial filing date prediction horizon and are therefore explored only at later administrative stages. Because these features are unavailable at the time of filing, they are treated as exploratory supplements rather than inputs to the primary opposition model.

The methodology proceeds in three stages:

1. **Transcription.** Audio recordings of City Council and Planning Commission hearings are transcribed locally using Whisper (faster-whisper). Each transcript is linked to its originating zoning case number, producing a corpus of 1,003 case-level text documents.
2. **Seed labeling via large language model.** A randomized sample of 50 transcripts is classified by a locally hosted language model (Llama 3.2, 3B parameters). For each transcript, the model is prompted to return two outputs:
 - **Stance:** Whether the overall testimony is supportive, opposed, or neutral toward the proposed zoning change.
 - **Argument frames:** Which of five predefined thematic categories appear in the testimony: (1) traffic and congestion, (2) infrastructure capacity, (3) displacement and affordability, (4) neighborhood character and compatibility, and (5) procedural fairness. These categories were selected based on recurring themes identified in the urban planning literature on participatory opposition.

The language model responses are parsed into structured binary labels (frame present or absent) and a three-class stance label. These 50 labeled cases serve as the training seed.

3. **Supervised expansion across the full corpus.** Using the 50 seed labels as training data, six logistic regression classifiers are trained—one for stance and one for each of the five argument frames—with TF-IDF unigram features (1,000-term vocabulary, English stop words removed) as inputs. Each classifier then predicts a continuous probability for every case in the full 1,003 transcript corpus. The result is a case-level matrix of six predicted probabilities: the

⁴Categories follow the 2024 City of Austin update. The original 2018 Uprooted report used a six-category gentrification typology (Susceptible, Early Type 1, Early Type 2, Dynamic, Late, Continued Loss) based on Bates (2013).

likelihood that the hearing testimony is oppositional, and the likelihood that each argument frame is present.

This approach is intentionally lightweight. The seed sample is small, at 50 cases, and the classifiers use simple linear models over bag-of-words features. Consequently, the predicted probabilities should be interpreted as coarse thematic indicators rather than precise measurements. The goal is to test whether even rough text signals carry discriminative power for opposition prediction at later horizons, not to build a comprehensive NLP model.

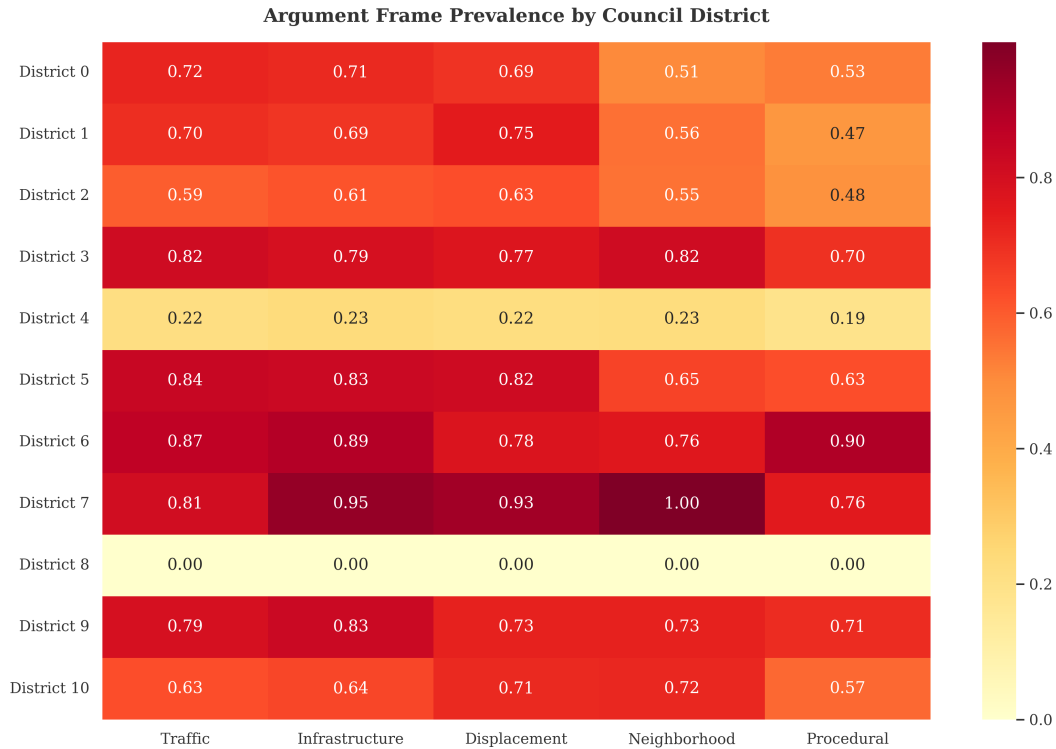


Figure 16: Mean predicted probability of each argument frame by Council District. Each of the 10 Austin Council Districts is shown as a row; columns represent the five argument frames classified from hearing transcripts. Darker cells indicate districts where a given frame appears more frequently in hearing testimony. Values are normalized to the maximum cell value for visual comparison.

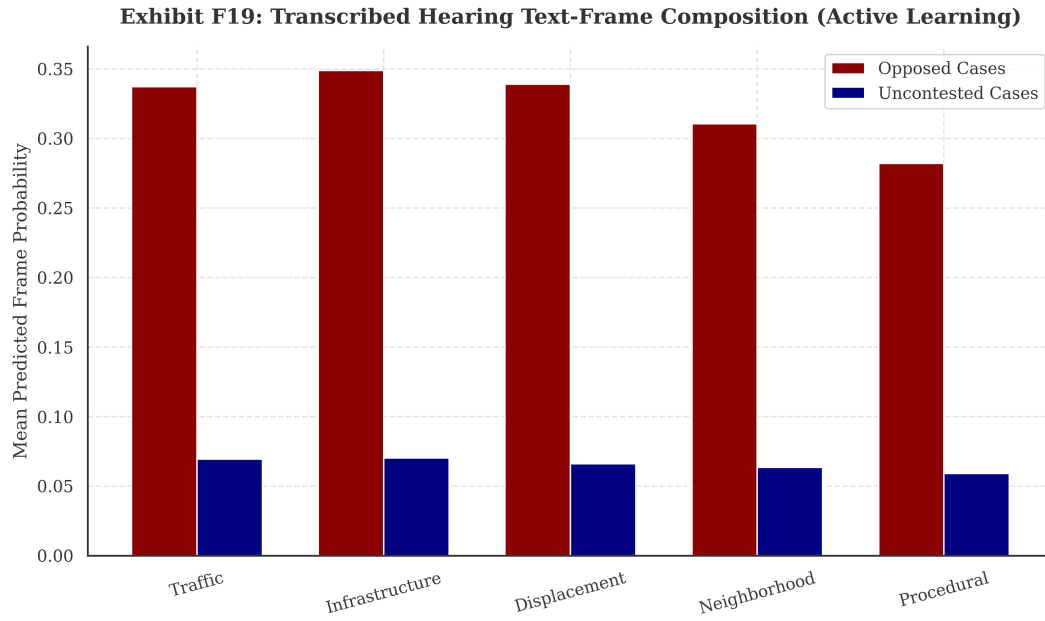


Figure 17: Mean predicted argument-frame probability for cases with a threshold-crossing petition filed (Opposed) versus cases without (Unopposed). Bars compare how frequently each thematic frame appears in hearing testimony across the two groups.

I Development Occurrence Model (Stage A)

This appendix reports the full Development Occurrence hazard model specification and evaluation. Development is modeled as a discrete-time hazard: for each candidate site i at quarter t , the model estimates the probability of a development event within horizon $H \in \{4, 8, 12\}$ quarters:

$$h_{i,t,H}^{dev} = P(\text{development event in next } H \text{ quarters} \mid X_{i,t}) \quad (2)$$

The unit of analysis is a candidate site-quarter, drawn from over 282,000 baseline Austin parcels aggregated to reflect contiguous configurations over time. Table 11 defines the nested target outcomes.

Table 11: Stage A Target Definitions

Target	Definition	Type
A1	Any housing production event (rezoning, permit, demolition)	Binary
A2	Discretionary zoning application subject to public hearing	Binary
A3	Net-new housing unit deliveries	Continuous

Because the baseline class imbalance is extreme—across 5,089,896 site-quarter observations, the empirical probability of a development event at any single site-quarter is approximately 2.5×10^{-5} —raw accuracy is uninformative given the extreme class imbalance. Table 12 reports hazard model performance relative to the baseline probability.

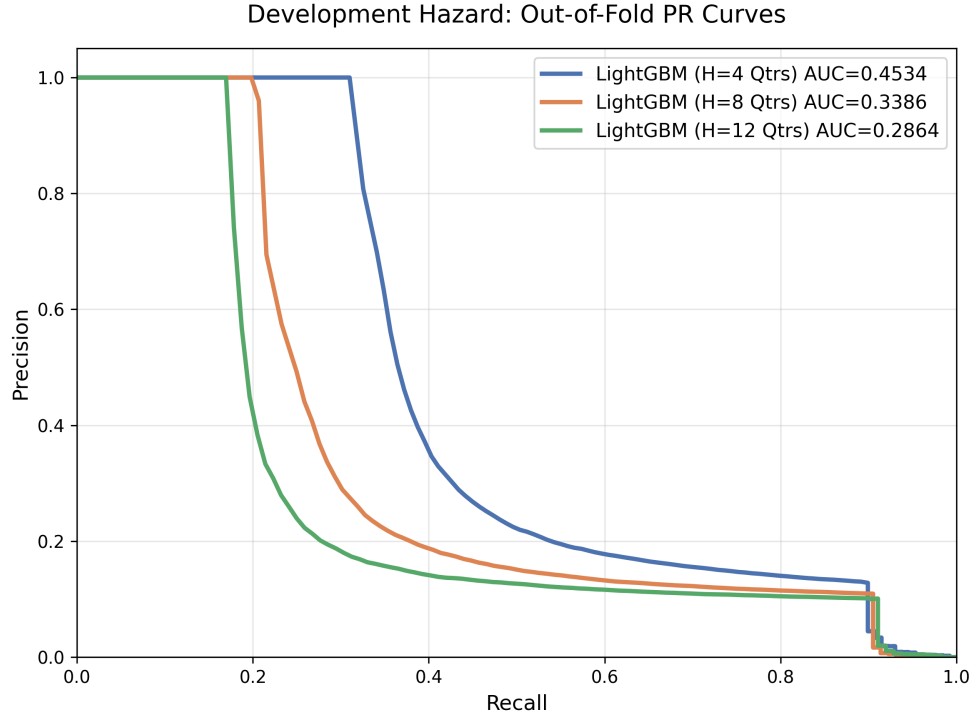


Figure 18: Precision–recall curves for the development occurrence hazard model across evaluation horizons.

Table 12: Development Occurrence Hazard Model Performance

Horizon	PR-AUC	Algorithm	Lift over Baseline [†]
4-Quarter	0.1917	LightGBM	$\sim 7,600\times$
8-Quarter	0.1201	LightGBM	$\sim 4,800\times$
12-Quarter	0.1127	LightGBM	$\sim 4,500\times$

[†]Lift defined as PR-AUC $\div$ base hazard probability ($\approx 2.5 \times 10^{-5}$). This is not a top-decile concentration lift; it reflects the extreme class imbalance of the site-quarter panel.

Multi-Horizon High-Probability Areas vs. Realized Development Events

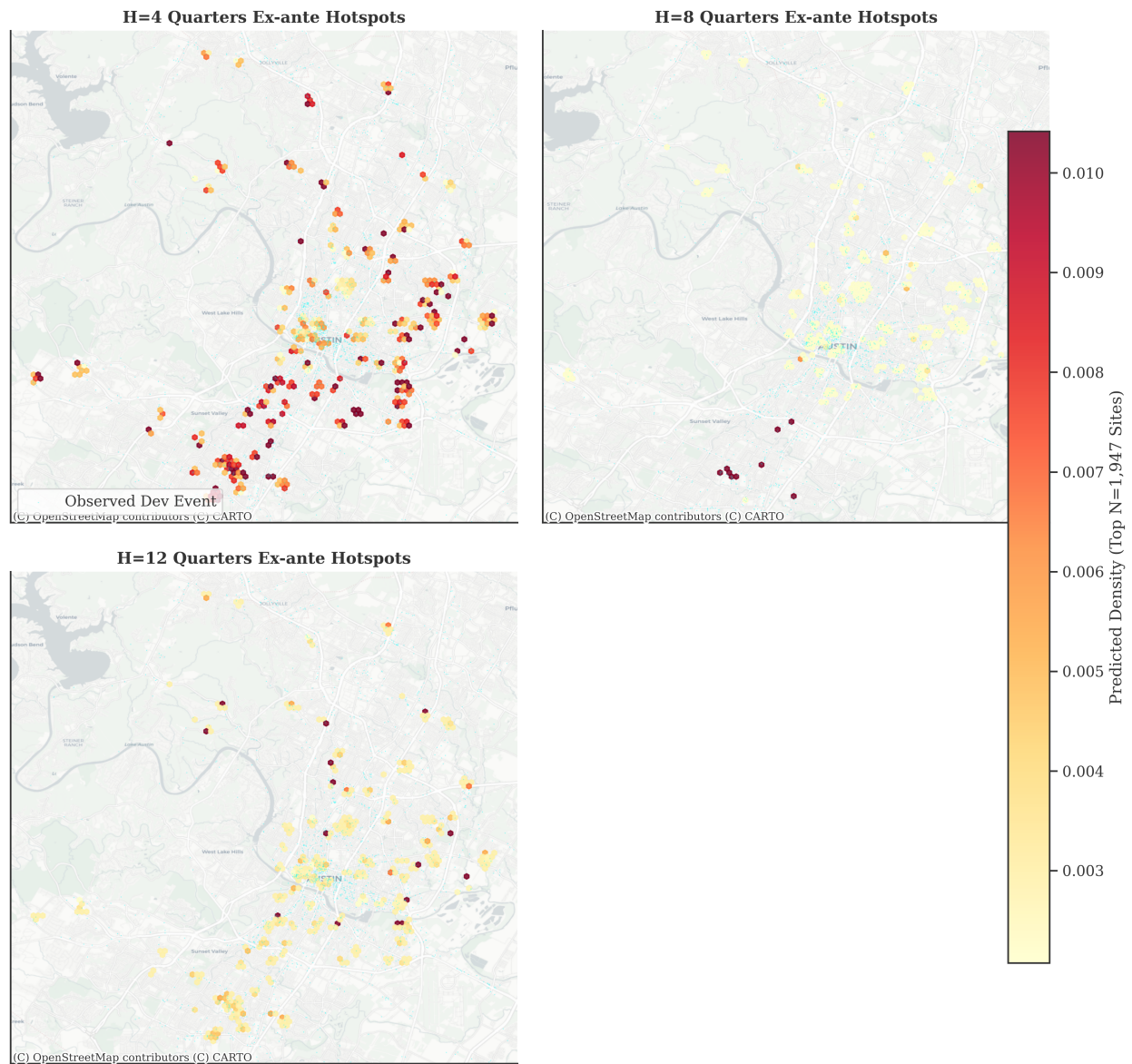


Figure 19: Ex-ante predicted areas of high predicted incidence density vs. realized development events. The map filters for the top decile of parcels with the highest predicted development hazard.

J Project Type and Scale Model (Stage B)

Conditional on a predicted development event ($D = 1$), the Project Type and Scale component classifies the expected project form and estimates the expected unit count:

$$E(\text{net new units}_{i,t,H} \mid D = 1, X_{i,t}) \quad (3)$$

The taxonomy classifies proposals into six categories defined by approval pathway and building form: PUD/Large Negotiated, Discretionary Rezoning, By-Right Infill, Missing-Middle, Mixed-Use, and Multifamily. Table 13 reports classification performance.

Table 13: Project Type Classification Performance. Macro-F1 scores calculate the unweighted simple average of F1 scores computed individually for each of the six project types (C). Defined as $\text{Macro-F1} = \frac{1}{|C|} \sum_{c \in C} F_{1,c}$.

Component	LightGBM	Random Forest	CatBoost	Logistic (L2)
<i>Macro-F1 Baseline</i>	0.748	0.739	0.548	0.177
<i>Class-Level Typology Performance (F1)</i>				
$\hookrightarrow$ PUD / Large Negotiated	1.000	1.000	1.000	0.097
$\hookrightarrow$ Discretionary Rezoning	0.850	0.863	0.724	0.455
$\hookrightarrow$ By-Right Infill	0.751	0.757	0.704	0.045
$\hookrightarrow$ Missing-Middle	0.519	0.507	0.392	0.257
$\hookrightarrow$ Mixed-Use	0.789	0.718	0.302	0.066
$\hookrightarrow$ Multifamily	0.579	0.590	0.165	0.140

K Political Outcome Model (Stage D)

Conditional on both a development proposal and the presence of a threshold-crossing petition, Stage D estimates the probability that a case survives to a final council reading rather than being withdrawn, stalled, or canceled before a vote occurs. Because virtually all cases that reach a final vote are approved, terminal council approval is near-deterministic and uninformative as a predictive target. Modeling pre-council attrition captures the institutional friction through which opposition most often affects outcomes.

Table 14: Pre-Council Attrition Forecast (Opposed Cases Only)

Model	Log Loss	Accuracy
LightGBM	0.0255	0.9976
CatBoost	0.1752	0.9440
Random Forest	0.2294	0.9854
Penalized Logistic (L2)	0.5951	0.3577

Accuracy and log loss are reported because the outcome (pre-council attrition/withdrawal) is highly imbalanced among opposed cases (77.6% attrition rate).

L Expected Contested Units (Composite Measure)

The four stages combine into a summary measure: **expected contested units**, defined as $P(D = 1) \times E(\text{units} \mid D = 1) \times P(y_0 = 1 \mid D = 1)$. This identifies locations where the city is most likely to encounter opposition to housing development. The multiplicative structure compounds uncertainty from each stage, so the resulting map should be interpreted as a relative ranking rather than a reliable absolute probability.

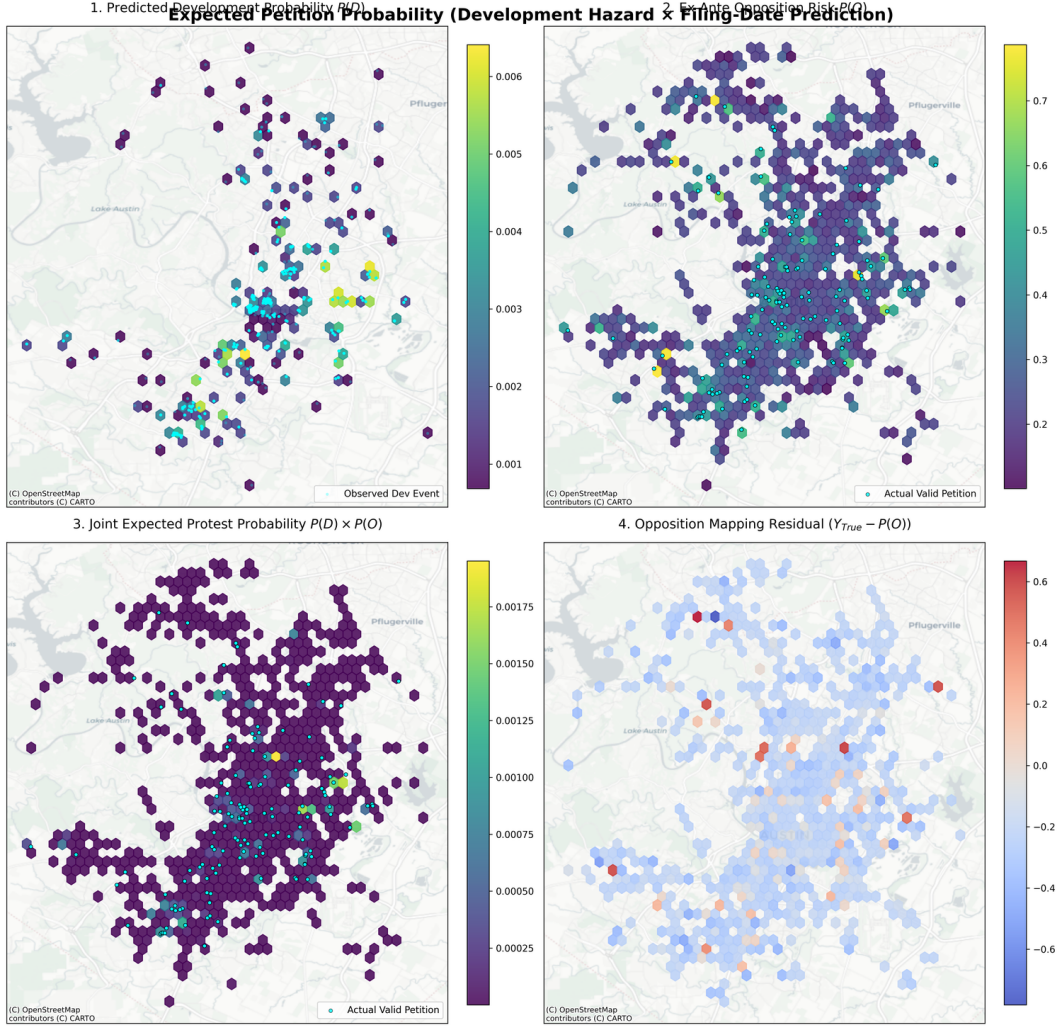


Figure 20: Expected units in projects facing threshold-crossing petitions (Stage A: H=4 Quarter Hazard $\times$ Stage C: filing-date prediction). Cyan dots in Panel A represent observed historical development zoning cases. Panels B and C highlight actual petition events mapped over the composite risk and prediction variance.

M Institutional Geographic Overlay Analyses

Additional analyses evaluate localized institutional overlays—Neighborhood Plan Areas (NPA), Historic District (HD) designations, Transit-Oriented Development (TOD) deregulation, and the 2015 transition to a 10-1 single-member council district system.

The ITS evaluation of the 10-1 transition yielded a coefficient of 0.119 on council dissent ($p = 0.344$), failing to reach statistical significance. The NPA overlay yielded a positive but insignificant coefficient of 0.239 ($p = 0.344$). Historic District and TOD overlays produced degenerate estimates with zero variance, likely because these zoning suffixes are rarely applied independently of major base-zone changes. These null and degenerate results are reported for transparency.

2022 Electoral Transition: Geographic Treatment Effects

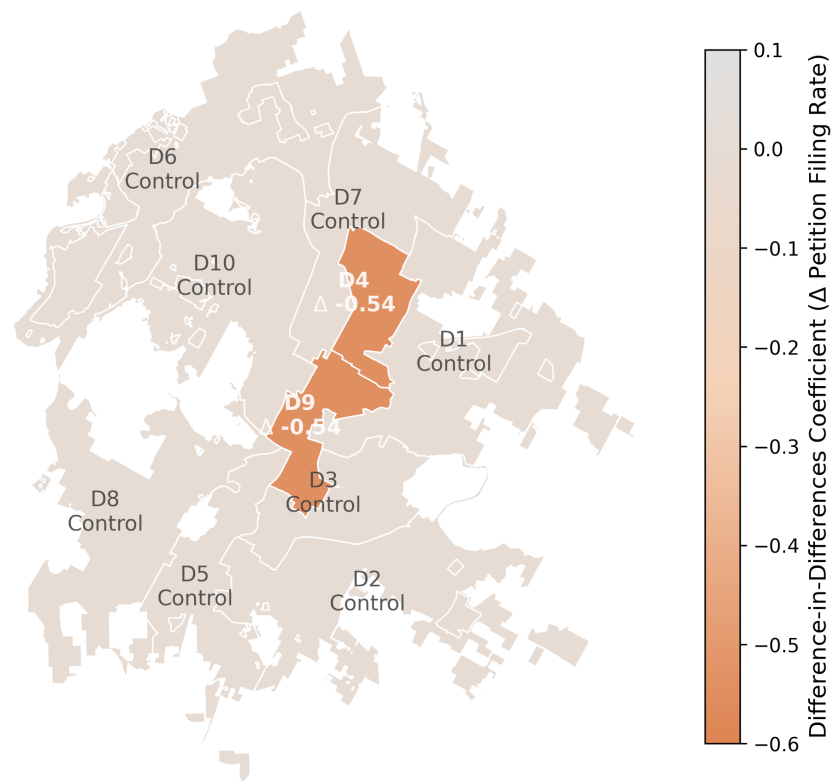


Figure 21: Geographic choropleth illustrating the descriptive interaction estimate. Districts 4 and 9 map the estimated coefficient ($\Delta - 0.546$), showing the interaction effect for districts that changed representation, relative to districts without comparable turnover.

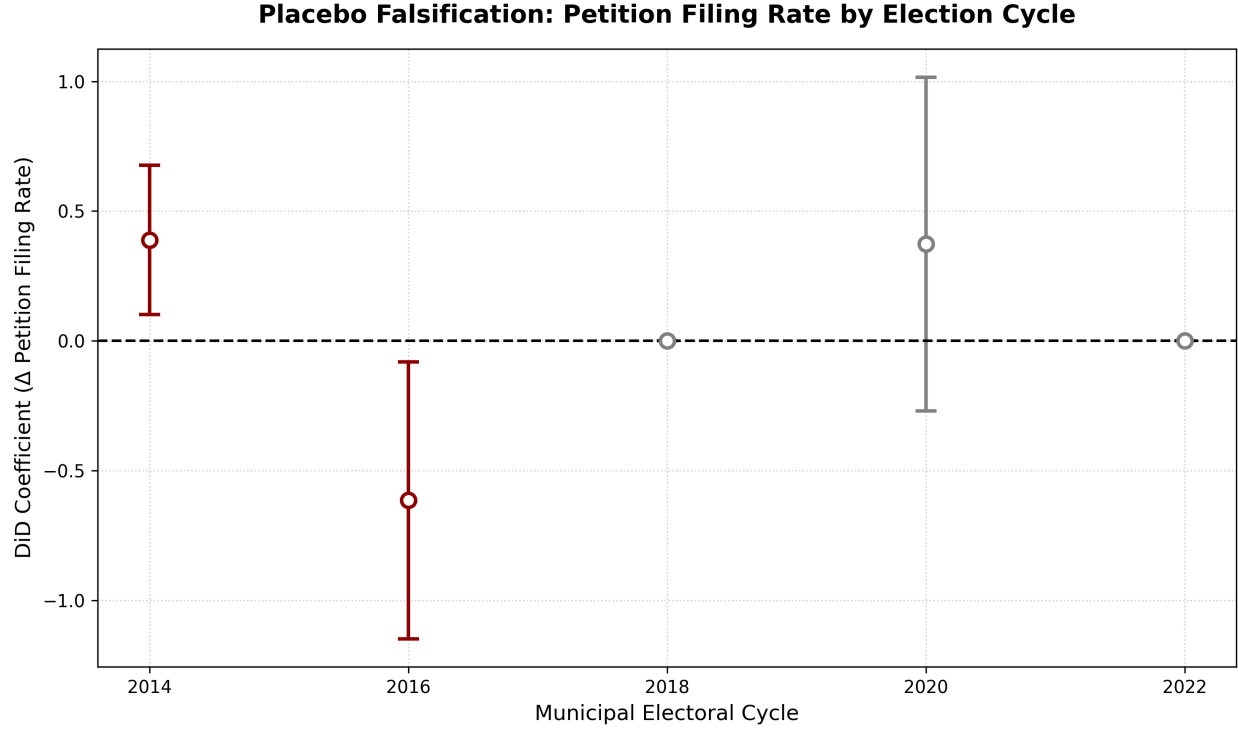


Figure 22: Placebo estimates across prior municipal electoral cycles (2014, 2016, 2018, 2020) for the affected districts. Only the 2022 cycle shows a detectable effect.

Table 15: Summary of Supplementary Quasi-Experimental Estimates

Design	Dep. Var.	Coeff.	SE	<i>p</i>	<i>N</i>
(1) 10-1 Council ITS	vote_no	0.119	0.122	0.344	20
(2) NPA Designation	vote_no	0.239	0.245	0.344	20
(3) Historic District	vote_no	0.000	—	nan	20
(4) TOD Deregulation	vote_no	0.000	—	nan	20
(5) 2022 Flipped District DiD	protested	-0.546	0.266	0.041	518

OLS with robust standard errors. Designs (3)–(4) are non-informative due to limited variation or zero-variance outcomes; they are reported for transparency rather than substantive interpretation. Design (5) reports the Flipped $\times$ Post-2022 interaction coefficient.

N HOME Initiative Event Study

Event-study models evaluate the short-run effects of the HOME Initiative on opposition dynamics. HOME Phase 1 and Phase 2 are treated as independent policy shocks, with treatment defined at the case level by parcel eligibility and timing anchored to application-acceptance start dates [5].

The event-study estimate for organized opposition, operationalized as the filing of a threshold-crossing petition per zoning case, is -0.177 (SE = 0.044, 95% CI: $[-0.26, -0.09]$). This estimate suggests a decrease in organized opposition, and the effect is statistically significant. The specification uses a Two-Way Fixed Effects (TWFE) model with case-level and time fixed effects, excluding post-treatment covariates to avoid “bad control” bias [10].

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension.

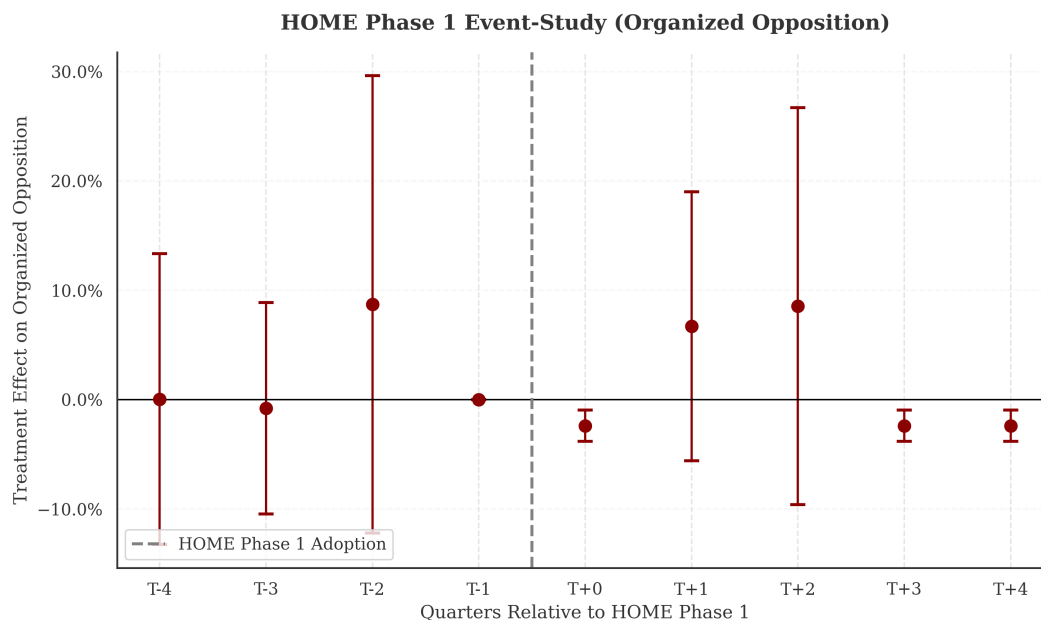


Figure 23: HOME Phase 1 event-study coefficients relative to the implementation date. The absence of a clean spatial control cohort necessitates purely temporal event-tracking.

O Invariant Causal Prediction (ICP)

As supplementary evidence on the causal structure of the feature space, an Invariant Causal Prediction (ICP) analysis [25] was conducted using 1,186 multi-parcel geometric environments. All 278 pairwise subsets of the 23 numeric features were rejected at $\alpha = 0.05$ after Bonferroni correction under a linear logistic specification, with a single marginal acceptance ($S = \{\text{land_to_total_ratio}, \text{year}\}$, $p_{\text{bonf}} = 0.051$). Nonlinear residual invariance tests (MLP and CVAE models under ERM and V-REx objectives) likewise universally rejected invariance. The tests consistently reject invariance, suggesting that the observable administrative feature space does not contain a stable set of causal predictors across environments for petition filing.